Internal Migration and the Spatial Reorganization of Agriculture *

Raahil Madhok†
University of Minnesota

Frederik Noack‡
University of British Columbia

Ahmed Mushfiq Mobarak§
Yale University

Olivier Deschenes¶
University of California, Santa Barbara

August 25, 2026

Abstract

How does migration out of rural areas reshape agriculture during structural transformation? Using household microdata from India, we show that agricultural households do not offset the loss of labor by increasing capital investment. Instead, they cultivate less land and lower technology use, leading to lower farm profits. Resulting changes in land and crop prices induce non-migrant agricultural households to expand farming operations. In aggregate, market adaptation mitigates 90% of the direct agricultural losses from urbanization. This prompts a spatial reorganization of agriculture: food production moves away from land near cities where emigration is high, toward remote areas where emigration is low.

Keywords: Internal migration, agriculture, structural transformation, India
JEL Codes: O13, O15, Q15, R11, J43

*We thank Werner Antweiler, Nathaniel Baum-Snow, Marc Bellemare, Francisco Costa, Bruno Esposito, Rick Hornbeck, Carol McAusland, and two anonymous referees for insightful comments. We also thank participants at the UCSD 21CIC Seminar, NEUDC, CREEA, UCSB Occasional Workshop, OSWEET, PacDev, AERE, Berkeley ARE, UMN Applied Economics, Aix-Marseille School of Economics, Yale Environmental Economics seminar and Paris Dauphine University. Noack thanks the Social Sciences and Humanities Research Council of Canada and the Canada Research Chairs Program for their financial support. All errors are our own.

†Department of Applied Economics, University of Minnesota; rmadhok@umn.edu
‡Food and Resource Economics Group, University of British Columbia; frederik.noack@ubc.ca
§Yale University and Deakin University; ahmed.mobarak@yale.edu
¶Department of Economics, UC Santa Barbara; olivier@econ.ucsb.edu

1 Introduction

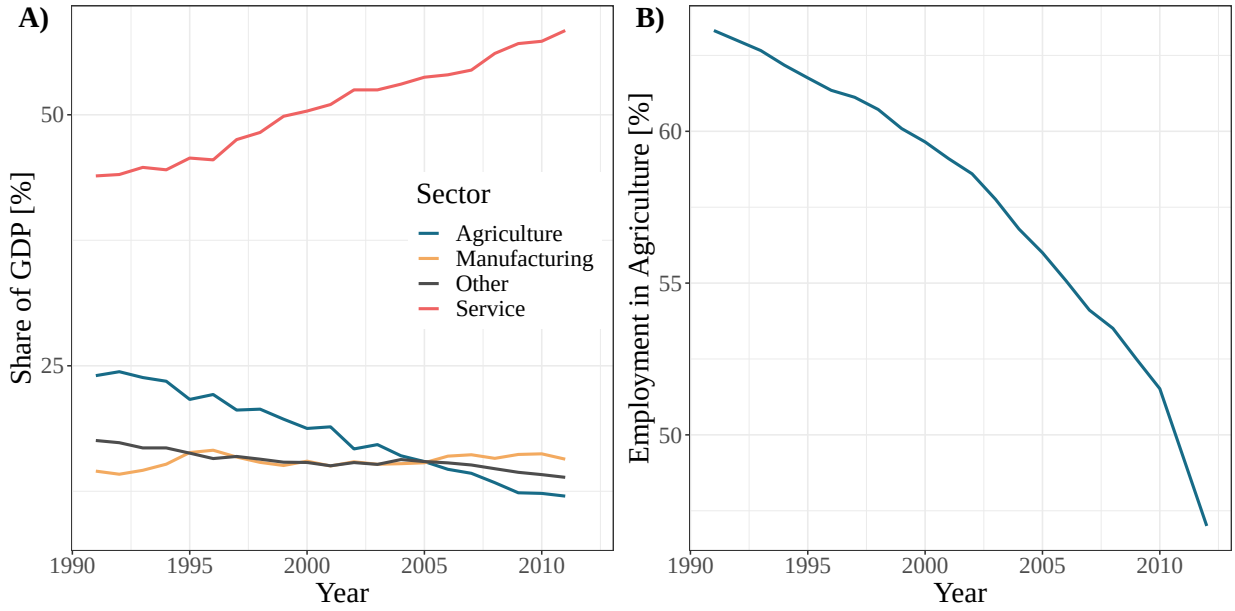
Internal labor migration is perhaps the most pervasive feature of structural transformation. America's rise in the early 20th century was powered by dramatic worker migrations from farms to cities in search of manufacturing and service jobs (Alvarez-Cuadrado and Poschke, 2011). China's 20th-century economic boom saw rural workers moving *en masse* to urban factories (Gao et al., 2022). Likewise, India's 1991 economic liberalization was followed by rapid growth in the service sector (Figure 1A, red) and massive exodus from agriculture, with the agricultural workforce shrinking by more than 30% over the next two decades (Figure 1B).

We use panel data on the migration and cultivation decisions of 42,000 Indian households during this period to re-visit foundational questions in development economics: do rural agricultural markets simply shed "surplus labor" à la Lewis (1954) without much production consequence, or does internal migration lead to a contraction of farming operations? What agricultural transformations occur to maintain domestic food production as labor exits agriculture? Is there agricultural modernization and mechanization, similar to what has been documented with county- and state-level data from the United States (Manuelli and Seshadri, 2014; Hornbeck and Naidu, 2014; Alvarez-Cuadrado and Poschke, 2011)?

Our first new insight based on the micro-data from India is that, contrary to the conventional wisdom developed in the historical US-based literature, farming households in India facing new migration opportunities *do not* mechanize; instead, they downsize. Our second insight is that agricultural transformation arises through a complex process in which factor markets readjust, and other non-migrant farming households in the same village or in other regions alter their farm operations in response to new factor prices, filling the agricultural production gaps. We design a credible shift-share instrument for migration and develop a research design to track indirect general equilibrium land and crop market adjustments. Unlike the Lewis (1954) view of surplus labor, our results imply that the marginal product of labor in agriculture is positive, and that adjustments occur across farms through crop and land markets, not necessarily within farms.

Our final insight is quantitative and descriptive. In aggregate, these direct and indirect effects reallocate agricultural activity away from rural areas near cities (where migration opportunities are plentiful), and toward more remote locations (where migration opportunities are scant). India's structural transformation was thus accompanied by a *spatial reorganization* of agriculture. We quantify the net effect on food supplies and find that indirect effects through land and crop markets recover roughly 90% of the direct loss in

Figure 1: Economic Growth and Reallocation of Labor (1991-2012)



Note: Panel A reports sectoral shares of India's GDP in 2005 prices using data from the Planning Commission of India. Panel B reports the percent of total employment in agriculture using data from ILOSTAT.

agricultural profits from households that emigrate and downsize their farms.[1]

The key challenge to identifying the effects of migration on agricultural development is that both processes are co-determined with origin-based push factors (Foster and Rosenzweig, 2007). We address this with a shift-share instrumental variable (IV) where the shift consists of income shocks at potential migration destinations. The share comprises two variables: inverse distance to these destinations combined with the household's migration potential. Since mostly men migrate for work in India, households with working-age men have more potential to react to destination wage changes. We thus measure migration potential by the number of working-age men at home during the baseline period. Interacting both share variables allows us to compare households with the same migration potential exposed to close and far income shocks as well as households exposed to the same income shocks but with high and low migration potential.[2]

[1]By aggregating our data from the household to the district level, we can replicate the mechanization results in the existing literature (Section 7.1). But our household data shows how and where this happens.

[2]We validate our empirical strategy using recent advances in the shift-share literature (Borusyak et al., 2025). The concern is that households with many working-aged men are systematically different than those with few. Borusyak et al. (2022a) show that shift-share instruments remain valid as long as the shift is quasi-randomly assigned. We provide two tests of this condition in Section 4.2.3. First, we recenter the instrument following Borusyak and Hull (2023), removing variation arising from systematic differences in expected shock exposure; estimates remain stable. Second, we present a two-period orthogonality test and find that the second period distance-weighted income shock is orthogonal to first period outcomes.

Another empirical challenge is that destination income growth can affect origin agriculture through channels other than migration. For example, urban growth may increase aggregate food demand or generate public investments that affect rural agriculture independently of any migration opportunities created in the process. Our reduced form estimates can thus be interpreted as the effects of nearby destination growth on origin agriculture through a combination of all these mechanisms. To isolate the migration channel, we control for inverse distance-weighted destination income changes—a lower-order component of our triple-interaction shift-share—in both our reduced form and 2SLS specifications to absorb destination-wide economic shocks that affect all households within an origin district. Identifying variation therefore comes from differential household exposure to common destination income shocks based on its migration capacity, rather than from aggregate demand spillovers. To isolate the effects of labor loss, we also report some specifications that separately control for remittance income that can attenuate the effect of labor loss by permitting households to invest in substitutes.

Our IV estimates show that migrant-sending households do not substitute either capital or hired labor for lost labor; they invest *less* in agricultural technologies, such as agrochemicals, irrigation water, and work animals, and less in non-family labor. This, in turn, drives a reduction in farm profits and farm size. Downsizing is mainly driven by larger farms becoming smaller, whereas smaller and medium-sized farms remain similarly sized. This is consistent with the notion that larger farms in India find it more difficult to hire non-family labor (Foster and Rosenzweig, 2022).

A general equilibrium model in the appendix guides our exploration of the indirect responses to labor migration by other households that have more limited exposure to urban opportunities. The model suggests that declining production and farm size in migrant-sending households would trigger higher crop prices and lower land prices. Remote households away from urban centers should respond by expanding farm activity, thus contributing to the structural transformation process indirectly. The key methodological insight from the model is that the direct effects of labor loss and the indirect effects through markets can be conveniently estimated via a single additive estimating equation. This approach draws on Adao et al. (2019) and Borusyak et al. (2022b), who develop this additive reduced form estimation equation from a theoretical gravity framework.[3]

In an expanded empirical framework that quantifies these direct and indirect effects together, we assume land markets operate at the village level: when some households

These tests imply that systematic differences between "exposed" and "non-exposed" households will not be picked up by the instrument since the shift is plausibly random with respect to baseline outcomes.

[3]See Huber (2023) for additional support for this empirical framework.

divest from agriculture as their prime-age males emigrate, local land prices fall, which enables non-migrant households in the village to expand production. Since India restricts cross-state agricultural trade, we conceive crop markets as areas *within a state* that are equally suitable for growing that set of crops. The crop market adjustment is measured as emigration among other households in those crop markets defined using gridded crop suitability indices. Intuitively, if urban shocks trigger mass emigration from a rice-suitable region, thereby reducing aggregate rice output and increasing rice prices, then non-migrant households elsewhere in the state will expand rice production if they also live in a rice-suitable region.

We build a district crop price index from ICRISAT and find that mass emigration from the crop market increases prices of those same crops, in line with our model. The price increase leads farming households with few or no migrants to invest more in agricultural technologies, such as agrochemicals and equipment rentals, ultimately leading to higher profits. On the land market side, we use land transaction data from IHDS and, as predicted by the model, find that emigration from the land market depresses land rents. This leads farmers remaining in those villages to exploit this opportunity and expand cultivation, again counteracting the downsizing of farms among migrant-sending households.

In this expanded framework, the direct effect remains instrumented with the shift-share IV, while indirect crop and land market adjustments are uninstrumented. This raises concerns because village-level shocks (e.g., drought, local agricultural policy) can co-determine village emigration as well as household agricultural decisions. Similarly, crop-specific shocks (e.g., crop disease, trade disruptions) can affect both crop-region emigration and household agriculture. We thus deploy a variety of robustness checks to support the credibility of our indirect crop and land market estimates. First, our estimates of indirect land market effects are robust when directly controlling for village-level shocks like droughts and agricultural extension activity. Second, we link each household's main crop to annual harvest prices and find highly stable estimates of crop market adjustments when controlling for crop price shocks. While this may appear to absorb the price mechanism itself, our identification strategy relies on deviations from *general* crop price trends induced by supply contractions due to aggregate emigration from the crop market. Third, we design placebo tests to validate our indirect estimates: household farm size does not respond to aggregate emigration by *non-agricultural* households in the village, suggesting that our land market measure is not picking up other non-land factor market changes. Households also do not respond to aggregate emigration from outside the crop market, reinforcing that our crop market measure captures adjustments specific to the relevant crop's supply region.

We find that the negative direct effect of migration on agricultural development dominates for households near cities that face low migration costs. The positive market-driven effects dominate for households further away who face higher migration costs. This leads to a spatial reorganization of agriculture toward remote areas. To test these spatial insights, we use geo-coded census data on economic growth and agricultural labor, as well as high-resolution satellite data on crop revenue from across India. At the district level, we show that agricultural labor declines near high-growth cities, while crop revenue rises in remote districts, mirroring our estimates of spatial reallocation through crop markets. Our high-resolution satellite maps show that crop revenue declines in peri-urban zones of a city with high labor outflows, and rise in the outer fringes, consistent with our estimates of spatial reallocation through local land markets. The consistent patterns we observe across both our econometric estimates and descriptive maps lend strong support to the idea that structural transformation reshapes the geography of agriculture.

The paper concludes with a simple accounting exercise based on our estimates to determine how much of aggregate migration-induced agricultural losses are mitigated by the crop and land market adjustments. We consider special cases of the empirical specification and predict aggregate crop value in the absence of migration, with migration but no market adjustments, with migration and land market but no crop market adjustments, and so on. We find that market adjustments mitigated 90% of aggregate agricultural losses due to emigration. These results do not necessarily imply that food production declined in India during this period of emigration and structural adjustment, only that we are able to quantify the partial effects of these two margins of adjustment.

1.1 Literature Contributions

This paper contributes to both the micro- and macro-development literature on structural transformation and its impact on the agricultural sector. A micro-development literature has studied a variety of agricultural responses to migration, including pollution (Garg et al., 2024), tree cover loss (Brewer et al., 2022), consumption volatility and crop choice (Kinnan et al., 2018), and technology choice (Mendola, 2008). Research in some developing contexts finds, like we do, that labor loss reduces cultivated area (Rozelle et al., 1999; Brewer et al., 2022), but others document land consolidation and mechanization with no loss to cultivation (De Janvry et al., 2015; Tian et al., 2023). Our contribution is to develop a unified framework to estimate both direct and indirect effects on agriculture and outline its spatial reorganization at the macro scale. Other studies have done this at small spatial scales (Blakeslee et al., 2023) or studied spillovers at different geographies separately

(Asher et al., 2023). We apply this framework in the case of India—the largest developing country—as it underwent a massive structural transformation.

The literature on the effects of international migration on families left behind (Dinkelman and Mariotti, 2016; Dinkelman et al., 2024; Khanna et al., 2026; Mobarak et al., 2023; Gibson et al., 2018) is also informative about the questions we pose. This literature highlights remittances sent back to finance domestic investments and education. As Mendola (2008) points out, remittances associated with international migration are much larger than those derived from internal migration, and are therefore used differently. Hossain and Mobarak (2026) explore the effects of loss in managerial capacity resulting from the emigration of a male household member from rural Bangladesh, which is distinct from the labor loss mechanism we highlight here.

We also contribute to a history literature that uses data at coarser spatial scales. Hornbeck and Naidu (2014) and Manuelli and Seshadri (2014) also study migration and agricultural development, but with county-level data from the early 1900s in the United States. They document a process of agricultural modernization whereby counties facing emigration shocks adopted new labor-replacing technologies. Similarly, Clemens et al. (2018) find that reduced agricultural immigration into the U.S. spurs adoption of labor-saving technologies. In these studies, technology adoption may be a direct response to labor loss or an indirect response to changing sectoral compositions across the broader economy. By separately quantifying direct and indirect effects, our micro data also allow us to paint a richer picture of the post-migration adjustment process.[4]

Lastly, our findings advance the modern macro development literature by elaborating the role of labor reallocation in driving agricultural development (Lewis, 1954; Harris and Todaro, 1970) using household microdata. A central finding of this literature is that large productivity gains arise as workers move from low-productivity agriculture to higher-productivity non-agricultural sectors (Restuccia et al., 2008; Duarte and Restuccia, 2010; Lagakos and Waugh, 2013; Gollin et al., 2014; Vollrath, 2009). Although this literature documents the aggregate importance of sectoral reallocation and the persistence of agricultural productivity gaps across countries, it provides limited evidence on how the agricultural sector itself adjusts as labor exits during structural transformation. We advance this work with household data to identify the micro-mechanisms. Our paper is especially related to Kaboski et al. (2024), who study direct and indirect effects of structural change on firms (rather than farms) in India.

[4] A related literature studies the effect of agricultural innovation on labor migration and finds that improvements in agricultural technology is labor-saving and leads to industrial growth (Caprettini and Voth, 2020; Bustos et al., 2016; Caunedo and Kala, 2021; Emerick, 2018; Moscona, 2019). We extend this literature by studying the reverse scenario and also by documenting market spillovers within the same sector.

2 Conceptual Framework

This section outlines a simple conceptual framework for understanding how destination income growth affects agricultural households. For intuition, we organize the framework around the case of rural-urban migration, although our empirical analysis allows for all types of internal migration flows, including rural-to-rural. The framework below highlights the main channels through which urbanization affects household crop production, investment, and land use – the three major outcomes we measure. We present a theoretical model that formalizes this discussion in Appendix C.

The first-order effect of urban growth is to raise the returns to non-agricultural employment. As urban growth pulls workers from nearby rural areas, the opportunity cost of farm labor increases. The ensuing labor loss through migration has ambiguous implications for agricultural production. If households can replace migrants with hired labor or labor-saving technology, then labor-capital substitution may maintain—or even increase—agricultural production, as documented in the historical United States (Hornbeck and Naidu, 2014). Alternatively, if households face market frictions, then lost labor cannot be fully replaced and agricultural production will decline. A third possibility is that rural labor is so abundant that its marginal product is near zero, in which case migration has no effect on agriculture (Lewis, 1954).

The main counterbalancing force operates through general equilibrium adjustments within crop and land markets. If urban growth leads to a contraction of agriculture, the declining food supply would raise crop prices, inducing other farming households to scale up production. This indirect crop market channel may be especially pronounced among farming households with few or no migrants, who avoid negative direct effects but still benefit from higher crop prices. Since these non-migrant households are more remote (Figure 2), we expect production booms in remote areas. Similarly, if the land that migrant households choose to cultivate shrinks, that would depress land (purchase or rental) prices, which may result in expansion of cultivation among other village residents through a land market channel.

While the above discussion emphasizes labor reallocation, there are several other ways that urban growth can affect rural agriculture independently of migration. Employment booms, construction activity, or improved credit access in cities can spill over to nearby origin districts by increasing aggregate demand, raising food prices, thereby changing rural household incentives for crop production. Migrants may also send remittances, which enables greater agricultural investment. Finally, improvements in market access associated with urbanization may further increase farm productivity.

Table 1: IHDS Summary Statistics

	IHDS-I (2004-05)		IHDS-II (2011-12)	
	Obs.	Mean	Obs.	Mean
A: Sampling Design				
Interviewed Households	41,554	-	34,643	-
Household Splits	-	-	5,375	-
Households Linked Across Waves	40,038	-	40,038	-
B: Household Migration				
Has Migrant	40,038	0.09	40,038	0.23
Has Working Age Male Migrant	40,038	0.07	40,038	0.17
Entire Family Moved in last 7 years	40,038	0.03	40,038	0.01
C: Household Agriculture				
Owns Land	40,017	0.48	40,038	0.46
Cultivated Area (ac.)	18,578	2.79	18,238	3.35
Total Labor Days (past year)	15,497	345.60	14,763	311.54
Agricultural Profits (Rs.)	19,202	14,938.77	16,306	16,765.44

Note: Data are at the household level. Household splits are households that split from the parent household after Wave 1. Working age spans 15-60. Total labor days includes hired and family labor. Profits are measured as farm income minus expenses, reported in 2005 prices.

In Section 5.1 we first estimate the reduced form relationship between urban growth and agricultural outcomes, which captures all channels jointly. We then isolate the migration channel by successively controlling for remittances, aggregate demand changes, and then ultimately instrument migration with a measure of household exposure to destination income shocks. As we will see, estimates remain remarkably stable, suggesting that labor reallocation is a key adjustment margin associated with urban growth.

3 Data and Background

We estimate the impact of internal migration on agricultural development using detailed household panel data. This section describes the data and presents three stylized facts about migration in India. These facts motivate the empirical strategy in Section 4.

3.1 Data Description

3.1.1 IHDS Panel Survey

Disaggregated household-level migration data are scarce in India. An exception is the IHDS panel survey, a nationally representative two-wave panel covering 384 districts (out of 594 at the time of the survey) across all states[5]. Table 1A describes the sample. Wave I (2004-05) surveyed 41,554 households. For Wave II (2011-2012), surveyors attempted to re-interview all Wave I households and also track households that split between rounds. They successfully recontacted 34,643 households, implying an attrition rate of 17%. In addition, 5,375 split-off households were linked back to their parent households, yielding a final panel of 40,038 households observed in both survey waves.

A key advantage of IHDS is that the same households are interviewed twice. This allows us to include household fixed effects in our regressions (Section 4.3) to control for time-invariant unobserved heterogeneity across households, such as caste or baseline poverty. Another advantage is that IHDS is one of the only Indian surveys with both a migration and farming module. We outline these data next.

3.1.2 Migration Data

In both survey rounds, enumerators ask respondents about migrant family members, defined as household members who have lived elsewhere *for over 6 months of the past year*. The IHDS definition thus characterizes longer-term migration spells[6], such as a son who has been away for the past five years. If he instead returned after five months, he would be a household resident, not a migrant. Survey respondents report the age, sex, and location of their migrant family members. We exclude international migrants ($<$ 10% of migrants), as our paper focuses on internal migration.

Table 1B summarizes household migration. The share of households with a migrant rises from 9% in Wave I to 23% in Wave II. Most migrants are working-age men aged 15-60.[7] Appendix Table A2 provides summary statistics at the migrant-level using the IHDS migrant roster. Migrants are predominantly male (79%) and migrate for jobs (68%), motivating our focus on working-age men as the key migrant demographic.[8] Most remaining

[5]Except Andaman and Nicobar Islands and Lakshadweep, which contain $<$ 1% of the population.

[6]Short-term ($<$6 months) circular migration is only documented in Wave II.

[7]To justify this age window, Table A1 reports migrant types by age group. Male migration for education sharply drops after age 14, which is when male migration for work jumps five-fold, suggesting a transition from school to work around age 15. The share of male labor migrants drops after age 59, suggesting they stop migrating around this age.

[8]Female migrants in IHDS do not generally reflect migration for marriage, as the IHDS migration module is not designed to capture marriage migration. In a patrilocal context, women who leave the

migrants leave for education and are excluded by our working-age window (Table A1). In terms of where they go, 65% of migration is within state. Figure B1 further decomposes migration streams by destination type and shows that rural-rural migration is the dominant form of within-state migration, while rural-urban migration accounts for the majority of cross-state flows. We, therefore, include both rural and urban destinations in the choice set when building our instrument for migration in Section 4.2.

3.1.3 Agriculture Data

Nearly half of the households in each IHDS round are engaged in agriculture (Table 1C). These households report agricultural capital, farm size, labor and profits. Agricultural capital is divided into input expenses and machinery ownership. Inputs include seeds, fertilizer, pesticides, irrigation water, and draft animals. Machinery includes tubewells, electric and diesel pumps, bullock carts, tractors, and threshers. To avoid multiple hypothesis testing, we aggregate the input and machinery variables into two separate indices following Anderson (2008). The approach combines the five underlying variables using inverse-covariance weights, which downweights redundant measures and upweights variables that contribute independent information. Unlike other popular methods such as principal components analysis (PCA), the Anderson indexing procedure accounts for correlation between the underlying variables.[9]

The farm labor module of IHDS records person-days of unpaid family labor as well as person-days and wages paid for hired labor in the past year. We measure labor demand as total person-days, which is the sum of family and hired labor. All monetary variables are deflated to 2005 prices using the rural or urban Consumer Price Index, depending on household location. Deflators are distributed as part of the IHDS data release.

Farm size is measured as the acreage of owned land that was cultivated by the household in the past year.[10] We focus on cultivation rather than ownership because that is the best representation of the agricultural margin of adjustment that we analyze. A decline in household cultivation could mean that the remaining land is left fallow, transitioned to

household upon marriage become members of their husband's household and therefore are not included in the migrant roster. The only exception is if they remain economically connected and regularly send money home. Non-resident married daughters—the survey's most closely related measure of marriage migration—account for only 1% of migrants in both survey waves.

[9] PCA finds the linear combination of variables that maximize the variance explained by the index. This can result in large weights for highly correlated variables as long as they account for a large share of the variation in the data. The Anderson index instead uses inverse-covariance weighting, which downweights redundant measures and gives higher weight to variables that contribute independent information.

[10] Wave I records total cultivated area in the past year. While Wave 2 does report cultivation separately by Kharif and Rabi season, we aggregate these to maintain consistency between survey rounds.

other land uses, or that it is rented or sharecropped out to other farming households. We explore these land use transitions in Section 6.6.

Lastly, we measure farm profits using the value computed by IHDS, which incorporates crop production and prices, farm input expenses, and agricultural rents paid and received. Thus, if migrant-sending households reduce cultivation but earn rental income by leasing land out, those rental receipts are already incorporated into farm profits. The profit measure is calculated as revenue net of expenses, which we deflate to 2005 prices. We cannot directly measure output changes since production quantities are only reported in Wave 2. For subsistence crops, profits are imputed based on prices that respondents report they *would have* received at market.

3.1.4 Covariates

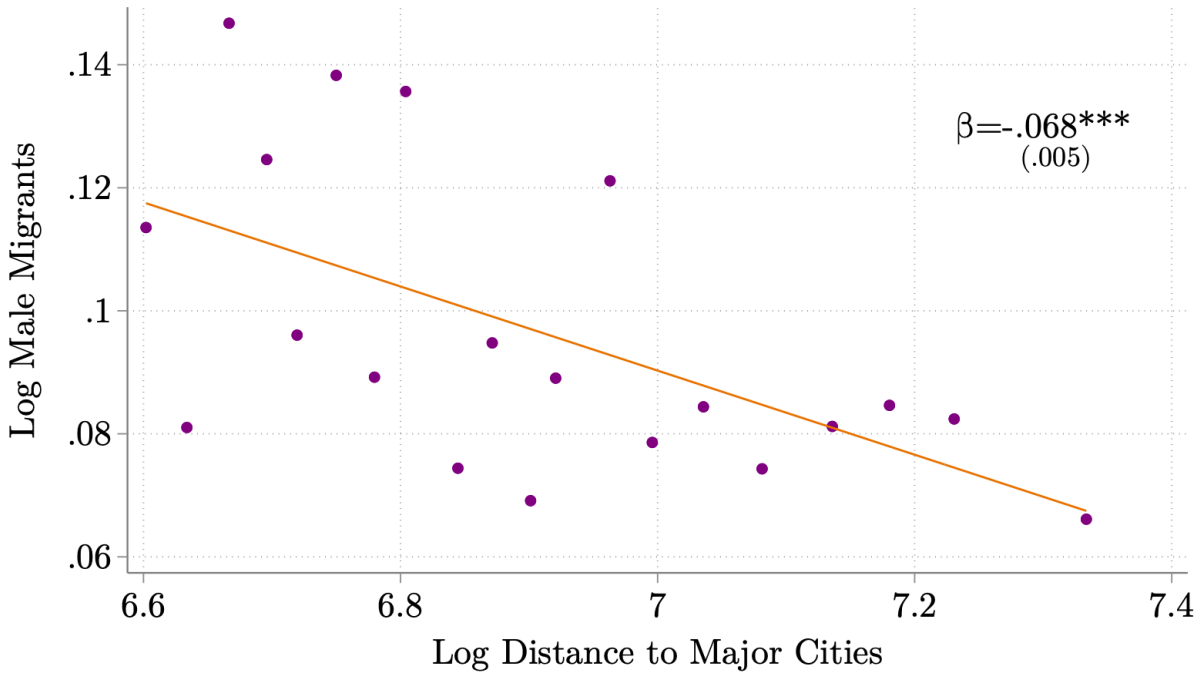
Droughts, temperature, and rainfall are the key covariates in our analysis. Accounting for climate is crucial because it affects both labor movement and agricultural decisions. We measure drought intensity using the gridded (0.5° resolution) Standardized Precipitation-Evapotranspiration Index (SPEI), which measures the difference between potential evapotranspiration and precipitation. Gridded annual temperature (°C) and rainfall (mm) are from the ERA5 product on a 0.125°× 0.125°grid (Hoffmann et al., 2019). For each covariate, we extract the mean over cells within districts and then compute annual averages for 2005 and 2012 to match with the IHDS.

3.2 Context on Agriculture and Migration in India

Next, we describe three stylized facts from the data that motivate our empirical strategy. The first fact is that entire-family migration is rare. Table 1B describes family migration between surveys, defined as households surveyed in Wave II that reported having moved after Wave I. Only 1% of households moved as a family (column 4). This motivates our investigation of how agricultural activities of household members remaining behind in origin areas are restructured in response to some family members migrating away.

The second fact is that remote households send fewer migrants. Figure 2 shows a binscatter plot of district remoteness, measured as mean distance to the 50 biggest cities, against number of household migrants, residualized on year fixed effects. The negative slope is the regression coefficient: on a log-log scale, households twice as remote send 7% fewer migrants, consistent with migration costs increasing with distance. This inverse correlation is robust when measuring remoteness as distance to the nearest large city. We make use of this fact to construct our instrument for migration in Section 4.2,

Figure 2: Individuals from remote places migrate less



Note: Y-axis is log number of working age male migrants in a household plus one, from IHDS. X-axis is log of mean distance from each district to the 50 biggest districts by population. Year fixed effects are partialled out. Orange line is the best linear fit, constructed from an OLS regression of y-residuals on x-residuals.

which leverages household exposure to destination income shocks where exposure declines with distance.

The third fact is that larger farms rely more on hired labor, which is consistent with previous work in India (Foster and Rosenzweig, 2022). Table A3 shows labor-to-land ratios and the hired labor share for small (0-2 acres), medium (2-4 acres), and large farms (4+ acres). While both family and hired labor intensity (person-days per acre) decline with farm size, family labor falls much more sharply. Consequently, although larger farms employ fewer hired workers per acre, hired labor accounts for nearly twice the share of total labor compared to small farms. This suggests that larger farms have less surplus family labor to buffer against labor loss and instead rely more on the market. As a result, migration may have larger production consequences on large farms, while also creating stronger incentives to substitute toward hired labor or capital. We examine this heterogeneity in Section 5.3.

4 Empirical Framework

This section develops an empirical strategy to estimate the *direct* effects of internal migration on agricultural development. Direct effects materialize through a labor-capital substitution channel, where crop and land prices are fixed. We develop a shift-share instrument for migration based on exposure to destination income shocks, and demonstrate the validity of our instrument through various validation exercises and tests suggested in the recent literature on shift-share designs. Section 6 extends this framework to investigate the *indirect* effects of migration through land and crop market adjustments.

4.1 Baseline OLS Equation

The relationship between agriculture and migration can be written as:

$$Y_{idt} = \beta_1 M_{idt}^{labor} + \Gamma X'_{dt} + \alpha_i + \gamma_t + \varepsilon_{idt}$$

where Y_{idt} are agricultural outcomes (e.g., household farm profits) for household i in district d at time t. M_{idt}^{labor} is the number of working-age male migrants sent from household i. X'_{dt} is a vector of covariates that jointly influence migration incentives and agricultural outcomes, such as weather conditions. Household fixed effects, α_i, absorb time-invariant characteristics of each household, such as distance to cities and land quality, that may affect both migration and agricultural decisions. Demand-side effects from increased urban productivity common to all households are captured by year fixed effects, γ_t.

β_1 captures the direct response of households to labor loss, and is likely biased when estimated via OLS due to the endogeneity of M_{idt}^{labor}. For example, while α_i absorbs baseline wealth, income changes could jointly influence migration and farming decisions. Improved agricultural technology can also release labor, leading to reverse causality. Our shift-share instrument, which we describe next, mitigates these threats to identification.

4.2 Shift-Share Instrument Design

4.2.1 Overview

We construct an instrument for household-member migration that combines plausibly exogenous destination income shocks with household-level exposure to these shocks. This setup is guided by a conceptual model presented in Appendix C. Destination income shocks include shocks to the service and manufacturing sectors (e.g., the rollout of telecoms) which are plausibly orthogonal to agricultural productivity in a given origin

location. We measure exposure to these shocks by (i) distance from the origin to potential destinations, which proxies for migration costs, and (ii) the baseline number of adult male household members, which captures the household's capacity for migration given that migration in India is predominantly male (Section 6.3).

Conditional on household and year fixed effects, the interaction of these two components generates variation in migration incentives that is independent of local or household-level characteristics. While household composition could influence agricultural investment directly, we show in the subsequent sections that household characteristics are uncorrelated with exposure to the shocks, ensuring that the instrument does not capture differential investment trends. Together with household and time fixed effects, this design thus exploits variation in both migration potential and migration costs across households and villages to identify the causal effect of migration.

4.2.2 Measurement

To identify β_1, we construct a shift-share instrument for M_{idt}^{labor} that combines income shocks at each potential destination (the shift) with measures of household exposure to the shocks (the share). When combined, the shift and share yield an instrument that isolates the pull stream of migration that is plausibly exogenous to origin push factors.

Income shocks, $inc_{d't}$, the "shift" of the shift-share design, are measured by mean household income in each potential destination district $d' \in \Theta/d$ at time t, where Θ is the set of all districts. We interact this with district population from the 2001 Census, $pop_{d'}$, to incorporate the fact that migration propensity increases toward urban destinations (Table A2). Because the income shock includes all potential destinations but places greater weight on more populous ones, we sometime refer to it as an "urban-oriented" income shock rather than a purely urban income shock.

Unlike previous studies, we use two variables to measure exposure to the shift. The first reflects the fact that potential migrants are more exposed to nearby shocks (Fact 2, Section 3.2). We measure this force by the inverse distance from each origin district d to every potential destination district $d' \in \Theta/d$, denoted as $\frac{1}{\tau_{dd'}}$, where τ is distance.[11] Since Θ spans all districts, households also consider rural destinations in their choice set, in line with the observed prevalence of rural-rural migration in India (Table A2). We set $\tau = 1$ based on similar values from the literature (Bryan and Morten, 2019; Schwartz, 1973), and consider alternative values as robustness checks (Section 5.4).

The second "share" of the shift-share instrument measures the number of baseline

[11] Distances are measured by kilometers between district centroids based on 2001 Census shapefiles.

working-age males living in household i, which we denote as φ_i. This term proxies each household's potential to benefit from destination income changes. We focus on males because internal migration in India is male-dominated (Section 3.1.2). As a robustness check in Section 5.4, we broaden this exposure measure to include both men and women.

Finally, we combine the shift and the shares to form an instrument for migration, z_{idt}:

$$z_{idt} = \underbrace{\varphi_i}_{\text{productivity (share I)}} \times \underbrace{\sum_{d' \in \Theta/d} \frac{1}{\tau_{dd'}}}_{\text{inverse distance (share II)}} \times \underbrace{(inc_{d't} \cdot pop_{d'})}_{\text{shift}} \tag{1}$$

4.2.3 Instrument Validity

z_{idt} is a valid instrument if: 1) it strongly predicts migration at the household level, M_{idt}^{labor}, and, 2) it satisfies the exclusion restriction. We test the first criterion with the first stage equation in Section 4.3. The exclusion restriction is that z_{idt} affects agricultural outcomes *only* through the migration of working-age male household members, conditional on household and year fixed effects and controls. While we cannot test this directly, we discuss potential violations through each component of z_{idt} and present robustness and sensitivity analyses in the sections below.

The first concern is that peri-urban households may face different input and output markets compared to more remote households, opening "backdoor" channels through which z_{idt} can affect household agricultural decisions (see Conceptual Framework). Likewise, urban income shocks can increase aggregate demand and, therefore, household crop production through output prices. We address these concerns in two ways: first, household fixed effects absorb time-invariant differences between households close and far from urban centers. Second, we control for the inverse-distance weighted income shock, $s_{dt} := \sum_{d' \in \Theta/d} \frac{inc_{d't} \cdot pop_{d'}}{\tau_{dd'}}$, in both the first and second stage. This term absorbs destination-wide economic shocks that affect all households within an origin district through aggregate demand, such that identification only relies on heterogeneous household *exposure* to the shock through differences in their productivity potential, φ_i.

The second concern is that households with more working-age males at baseline, our measure of φ_i, may be on different agricultural development trajectories than those with fewer working-age males. We address this in several ways: first, we show in Section 5.1 that our reduced form estimates are robust to allowing male-dominated households to follow separate time trends. Second, we follow recent work showing that shift-share instruments with endogenous shares are valid if the shock is as-good-as-randomly assigned (Borusyak et al., 2022a, 2025). Figure B3A shows that the histogram of the shock, s_{dt}, is

Table 2: Orthogonality Between Second Period Shocks and First Period Outcomes

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Distance Wt. Income (t=2012)	0.082	0.066	-0.084*	0.064	-0.020
	(0.093)	(0.052)	(0.048)	(0.074)	(0.066)
Household Males	Yes	Yes	Yes	Yes	Yes
Origin Income	Yes	Yes	Yes	Yes	Yes
State FEs	✓	✓	✓	✓	✓
Observations	16874	15812	18213	15208	18822
R^2	0.275	0.209	0.204	0.244	0.142

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Data are a cross section of households. Outcomes are for 2005 and are in standardized units. The dependent variables are as follows: Column 1 is an index of agricultural technology expenditures, and Column 2 is an index of machinery ownership. Column 3 is farm size, measured as cultivated acres. Column 4 is total labor person-days of hired and family labor. Column 5 is farm profit, defined as farm revenue minus expenses. All regressions include state fixed effects and control for district income and number of household working-age males in 2005. Standard errors clustered by state.

nearly identical among households with high and low values of φ_i. This means that both groups are similarly exposed to the shock, and that the shock is thus unlikely to pick up differential trends of male-dominated households. In Section 5.4, we generalize this test by re-centering the instrument following Borusyak and Hull (2023).

Lastly, we conduct a lead test with our two-period panel by regressing a set of baseline period agricultural outcomes, Y_{idt_1}, on the second-period shock, s_{dt_2}, in a pooled cross-section:

$$Y_{idt_1} = \phi \cdot s_{dt_2} + \beta \cdot \varphi_i + \Gamma X'_{dt_2} + \gamma_s + \varepsilon_{idt} \tag{2}$$

where ϕ is the balance coefficient of interest. Controlling for φ_i ensures that we isolate the relationship between future destination income shocks and baseline agricultural outcomes, holding migration capacity constant. If s_{dt_2} is as-good-as-randomly assigned, it should not predict Y_{idt_1} and thus $\phi = 0$. X'_{dt_2} is a covariate vector including district income and γ_s is a state fixed effect. Table 2 reports estimates of ϕ from regressions with agricultural technology, farm size, labor-days, and profits as outcomes. The second-period distance-weighted income shock, s_{dt_2}, is uncorrelated with baseline agricultural technology, labor, and profits. Farm size exhibits a small and weak negative coefficient ($p = 0.095$). Taken together, these results suggest that the destination income shock is unlikely to capture systematic differences in pre-existing agricultural characteristics or trends among households with different migration capacity (Borusyak et al., 2022a,

2025). It also helps rule out anticipation effects: if households reduce investment because they expect future urban opportunities to make farming less worthwhile today, then ϕ would have been negative and significant. The absence of a relationship thus suggests that our main estimates (Section 5) reflect direct responses to labor reallocation rather than forward-looking reductions in agricultural investment.

Despite these checks, the exclusion restriction ultimately requires that destination income shocks affect agricultural outcomes through migration rather than through other channels. Readers who remain concerned about this assumption can instead interpret our reduced-form estimates in Section 5.1, which do not require taking a stance on the specific mechanism linking destination opportunities to household agricultural outcomes. These estimates are, unsurprisingly, similar to the 2SLS estimates. Under this more conservative interpretation, our results capture the effect of destination income growth—which affects migration patterns but may also influence rural economies through other channels—on the spatial reorganization of agriculture.

4.3 Two-Stage Least Squares

Equipped with our shift-share instrument, z_{idt}, and having shown evidence supporting its validity, we proceed to estimate the effect of labor migration on agricultural outcomes in a 2SLS framework as follows:

$$M_{idt}^{labor} = \mu_1 z_{idt} + \mu_2 s_{dt} + \mu_3 inc_{dt} + \Gamma X'_{dt} + \alpha_i + \gamma_t + \varepsilon_{ijdt} \tag{3}$$

$$Y_{idt} = \beta_1 \widehat{M_{idt}^{labor}} + \beta_2 s_{dt} + \beta_3 inc_{dt} + \Gamma X'_{dt} + \alpha_i + \gamma_t + \eta_{ijdt} \tag{4}$$

where all terms have been defined in the previous subsections. Equation 3 is the first stage, which relates labor outflows from the origin, M_{idt}^{labor}, to the instrument, z_{idt}, conditional on the distance-weighted income shock itself, s_{dt}. We control for per-capita incomes in the origin district, inc_{dt}, to account for the possible correlation between destination and origin income shocks. X'_{dt} is a vector of covariates including drought intensity, temperature, and rainfall. α_i and γ_t are household and year fixed effects, respectively. To the extent that z_{idt} is plausibly exogenous (Section 4.2.3) conditional on fixed effects and controls, $\widehat{M_{idt}^{labor}}$ represents the pull stream of migration that is orthogonal to push incentives at the origin, and μ_1 identifies the migration response to destination income shocks.

Equation 4 estimates the second stage impact of migration on agricultural activity, Y_{idt}, of household i in district d at time t. The main outcomes are technology adoption, farm size, labor-days, and farm profits. The coefficient of interest is β_1, which captures

the direct effect of emigration on agricultural development through the labor channel. $\beta_1 > 0$ indicates that households respond to migration-induced labor loss by expanding their farm activity, in line with the idea of labor-capital substitution. $\beta_1 < 0$ implies that households downsize their farming operations.

Since s_{dt} is a component of z_{idt}, and also included directly as a control, β_1 is identified off of differences in household i's exposure to *changes* in destination incomes, where exposure declines with distance and increases in the number of resident working-age males. Inclusion of origin district income, inc_{dt}, as a control ensures that identifying variation captures the bilateral nature of migration decisions whereby migrants respond to *differences* between origin and potential destination incomes (Borusyak et al., 2022b).

5 Results: The Direct Effect of Migration on Agriculture

We find that Indian households do not replace labor lost to migration with capital. Instead, they *reduce* farm size and agricultural investments in response to labor migration. Although crop and land prices are assumed to be fixed here, Section 6.2 extends the analysis to enable crop and land market adjustments.

5.1 Agricultural Responses to Destination Income Growth

Before presenting the main results, we first present reduced-form estimates directly linking agricultural outcomes to the instrument that measures rural household exposure to the urban income shock. Urban income growth can affect nearby rural districts in multiple ways: attracting migrant workers, but also from demand shocks related to employment booms or infrastructure upgrades, or remittance transfers to finance agricultural investments. These mechanisms are captured by the lower-order components of our triple-interaction instrument (Equation 1). We therefore successively control for each lower-order term in the different columns of Table 3, before isolating variation most plausibly related to the labor migration mechanism in the saturated specification. We hold origin incomes constant throughout so that identification comes from differences between destination opportunities and those at home. We use farm profits as the outcome in this table, and investigate capital, labor, and land adjustments in subsequent sections.

Column 1 shows reduced form estimates without controlling for lower-order components of the shift-share. The negative coefficient thus captures the *total* effect of destination income growth on farm profits. Column 2 adds a control for distance-weighted destination incomes, s_{dt}, a key lower-order term that absorbs destination-wide economic

Table 3: Reduced Form—Agricultural Responses to Destination Income Growth

	(1) Farm Profit	(2) Farm Profit	(3) Farm Profit	(4) Farm Profit
z_{idt}: Wt. Income × Baseline Males	-0.752***	-0.776***	-0.774***	-0.629***
	(0.114)	(0.115)	(0.115)	(0.171)
Origin Income (inc_{dt})	Yes	Yes	Yes	Yes
Wt. Income (s_{dt})	No	Yes	Yes	Yes
Remittances	No	No	Yes	Yes
Male Time Trends	No	No	No	Yes
Outcome SD	26744.428	26744.428	26744.428	26744.428
Explanatory SD	8.563	8.563	8.563	8.563
HH FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	27462	27462	27462	27462

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Outcomes and all explanatory variables are standardized to have a mean of 0 and a standard deviation of 1 (the actual standard deviations of each variables are reported as "Outcome SD" and "Explanatory SD") . "Wt. Income" is inverse-distance, population-weighted income (s_{dt}). "Baseline Males" is number of baseline working age male residents (φ_i). "Origin Income" (inc_{dt}), is district-level mean per capita income. Male time trends are above-median baseline males × year fixed effects. Farm profits are defined as farm income minus expenses. All specifications control for drought, temperature, and rainfall. Standard errors clustered by district.

shocks that affect all households within an origin district through aggregate demand. The coefficient becomes marginally more negative, suggesting that positive demand spillovers partly offset the adverse effects of destination income growth in column 1. The specification in column 3 adds a control for remittances, which captures household-specific transfers not already accounted for by the aggregate demand term. The coefficient remains essentially unchanged.[12]

At this point, the identifying variation comes from differential household exposure to a common destination income shock based on baseline migration capacity. Column 4 addresses the concern that the number of baseline working-age males may proxy for household characteristics associated with differential agricultural trends unrelated to migration. We therefore allow households with above-median baseline numbers of working-age males to follow different time trends.[13] The point estimate becomes slightly smaller, but the change is small relative to the standard errors, suggesting that differential agri-

[12]We note that the other two-way interaction between baseline males and destination incomes cannot be separately identified because households are simultaneously exposed to income shocks from many potential destinations. This two-way interaction only becomes meaningful after aggregating destination shocks using the inverse-distance weights, at which point it becomes the full triple-interaction shift-share.

[13]We use above-median instead of the continuous measure of baseline males in the interaction because the latter is already part of the instrument, which substantially weakens the first stage. Using an above-median indicator provides a flexible test for differential trends while preserving identifying variation.

Table 4: IV Estimates—Direct Effect of Migration on Agricultural Development

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants (σ)	-1.081***	-0.737***	-1.238***	-1.913***	-0.978***
	(0.239)	(0.222)	(0.266)	(0.333)	(0.199)
Wt. Income (s_{dt})	Yes	Yes	Yes	Yes	Yes
Origin Income (inc_{dt})	Yes	Yes	Yes	Yes	Yes
Outcome SD	-	-	3.621	211.329	26744.428
Explanatory SD	0.524	0.521	0.533	0.518	0.525
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	25928	24970	29346	20910	27462
F-Statistic	54.4	55.1	46.3	52.1	59.4

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Outcomes and all explanatory variables are standardized to have a mean of 0 and a standard deviation of 1 (the actual standard deviations of each variable are reported as "Outcome SD" and "Explanatory SD"). Male Migrants is the number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with the number of baseline working age males in the household. "Wt. Income" is inverse-distance, population-weighted income (s_{dt}). "Origin Income" (inc_{dt}), is district-level mean per capita income. Columns 1 and 2 are indices for agricultural expenses and machinery, respectively (Section 6.1). Column 3 is farm size, measured as cultivated acres. Column 4 is labor, measured as total person-days of labor (family + hired workers). Farm profit (column 5) is defined as farm income minus expenses. All specifications control for drought, temperature, and rainfall. Standard errors clustered by district.

cultural trends associated with baseline male composition are not a major confounder. Combined with our instrument validity tests in Section 4.2.3, these results support the interpretation that the remaining identifying variation reflects differential household exposure to common destination income shocks based on its migration capacity, rather than from aggregate demand spillovers or other confounding channels.

5.2 Migration leads to agricultural decline

Table A4 presents the first stage estimates of Equation 3. Column 1 excludes controls, column 2 controls for the direct shock, s_{dt}, and column 3—our preferred specification—controls for both s_{dt} and origin income, inc_{dt}. We do not control for remittances because they are a post-treatment outcome of migration. The instrument strongly predicts migration-induced labor outflows across all specifications. To interpret the coefficient, note that the average household has 1.76 male working-age residents. Therefore, a 1σ increase in destination incomes pulls 20% (=0.356/1.76) of them away to join the destination

labor force. F-statistics are well above rule-of-thumb levels in tests for weak instruments (Kleibergen and Paap, 2006).

Table 4 presents second stage estimates of Equation 4. All variables are reported in standard deviation units to help compare coefficient magnitudes across outcomes. The coefficient of interest is negative and statistically significant for all outcomes, indicating that labor loss through migration prompts agricultural decline within the household, not modernization.[14] The point estimate in column 1 implies that a 1σ reduction in farm labor causes households to reduce agricultural technology expenses by 1.08σ. When decomposing the index by technology type (Table A7, columns 1-5), the effect is primarily driven by lower spending on seeds, agrochemicals, irrigation water, and rented equipment. Column 2 of Table 4 shows that labor loss also causes households to reduce machinery holdings. This is driven by less investment in tubewells, water pumps, bull carts, and threshers (Table A7, columns 6-10).

Columns 3-4 of Table 4 explore additional response margins. Column 3 shows that labor loss reduces cultivated area by 1.24σ, in line with the idea that declining labor and capital reduce the marginal productivity of land, which in turn prompts farm contraction (Appendix C). As we show in Section 5.3, the large coefficient is driven by large farms downsizing, whereas small and medium farms remain of similar size. When restricting the sample to farms < 10 acres, the effect size is much smaller: -0.34σ. The distribution of farm size in our data has a few unusually large outlier farms (Figure B2), which account for the unusually large effect size on downsizing in column 3. The -0.34σ effect after outliers are eliminated is more representative.

Column 4 shows that total labor days, including family and hired labor, declines in response to emigration of household members. This is consistent with imperfect rural labor markets, since households do not compensate for the loss of family labor by hiring more workers. A 1σ increase in migration leads to a 1.9σ reduction in farm labor, corresponding to roughly 400 person-days. Lower investment in capital, land, and labor leads to lower farm profits (column 5). The point estimate implies that a 1σ increase in migration leads to a 0.98σ decline in farm profits over the past year, corresponding to about Rs. 26,000.

Finally, since labor (column 4) declines more than capital (columns 1 and 2), agriculture may become more capital-intensive. We test this using the capital-labor ratio (total

[14]The corresponding OLS estimates in Table A5 are positive, small, and relatively noisy. This is likely because migration is endogenous due to two counteracting forces: on one hand, richer households can send more migrants and also have access to better technology, leading to a positive correlation between migration and agricultural development. On the other hand, urban opportunities draw workers out of agriculture, leading to agricultural contraction (Table A4, Table 3). These push and pull factors draw the coefficient in opposing directions, overall leading to attenuated OLS estimates. Our IV estimates overcome this issue by isolating the pull stream of migration that is plausibly orthogonal to push factors at the origin.

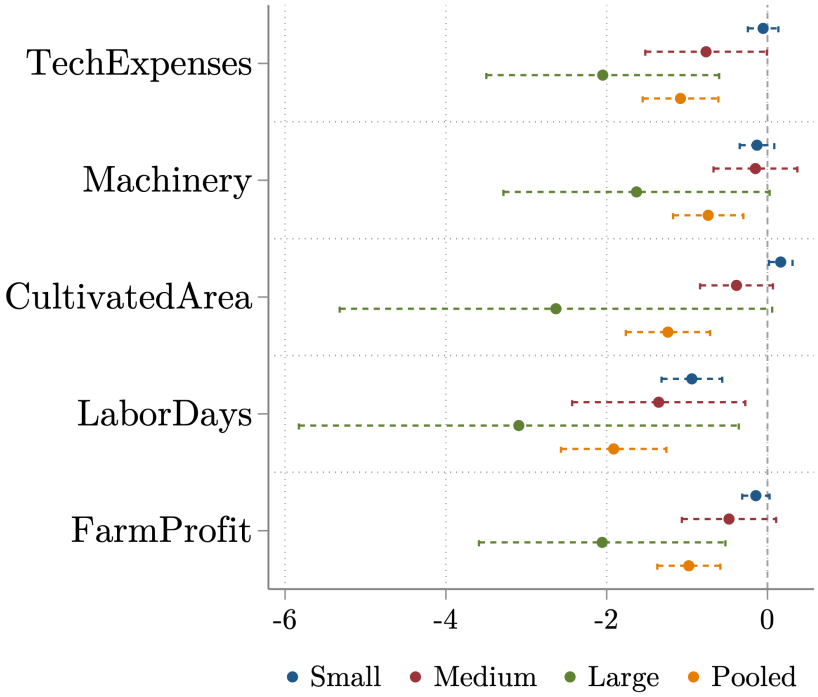


Figure 3: IV Estimates—Direct Effects of Migration on Agriculture by Farm Size

Note: The explanatory variable is the number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with the number of baseline working age males in the household. Outcomes (on the y-axis) and the instrument are in standardized units. Blue, red, green, and orange coefficients are for small (0-2 ac.), medium (2-4 ac.), large (4+ ac.), and pooled farm sizes, respectively. Tech Expenses and Machinery are indices for agricultural expenses and stock of machinery, respectively (Section 6.1). Farm size is measured as cultivated acres. Labor is person-days of labor (family + hired). Farm profit is defined as farm income minus expenses. All specifications control for drought, temperature, and rainfall. Standard errors clustered by district.

agricultural capital expenses divided by total labor days) as an outcome. The estimates in Table A6 are consistent with capital intensification, although they are not statistically significant in the IV specification. We do, however, find evidence of capital uptake in the analysis at the district-level (Section 7.1) when general equilibrium adjustments operate.

5.3 Estimates by Farm Size

To test whether our results in Table 4 are a direct response to labor loss, we compare responses of labor-constrained farms, where farmers are sensitive to labor loss, to surplus-labor farms, where the loss of a marginal worker should have less impact. Our setting is well suited for this test since, as documented in Fact 3 (Section 3.2), labor constraints increase in farm size. We therefore expect agricultural responses to migration-induced

labor loss to increase in farm size as labor constraints become more binding.

Figure 3 estimates Equation 4 by farm size. Although labor use declines across all size groups, large farms (green) are the ones that contract their operations the most. The wider confidence intervals are due to having fewer large farms in the sample.[15] Heterogeneity therefore appears in how farms adjust to labor loss, rather than in the labor loss itself. This is consistent with Fact 3 (Section 3.2): small and medium farms rely more on family labor, whereas hired labor dominates on large farms. Thus, while farms of all sizes reduce labor, the broader contraction of inputs and machinery is concentrated on large farms, consistent with their having less family labor to buffer the loss of a household member.

Note that farm size captures other differences that may determine adjustment to labor loss; for example large farms may be more mechanized at baseline, enabling greater scope for downscaling compared to smaller farms. However, small and medium farms in our sample also use substantial inputs, leaving ample scope for adjustment in response to migration. Yet we observe little such adjustment. We thus interpret the stronger response of large farms as consistent with tighter labor constraints, while recognizing that other differences across farm size may also contribute to the observed heterogeneity.

5.4 Robustness Checks

To probe the robustness of our IV estimates, we first address potential nonrandom exposure to destination income shocks by recentering our instrument following Borusyak and Hull (2023). We reshuffle population-weighted income shocks ($inc_{d't} \cdot pop_{d'}$ in Equation 1) across districts 1000 times, reconstruct the shift-share each time, and average across iterations to obtain the expected instrument. We then include this as a control in Table A8, which removes the part of the shift-share that reflects a district's tendency to be more or less exposed regardless of which shock actually occurs. The estimates remain stable, reinforcing the evidence from Section 4.2.3 that systematic differences in household exposure to the shocks do not drive our results.

Next, we implement a series of sensitivity tests with alternative fixed effects, controls, shift-share designs, and distance decay functions. Panel A of Table A9 tests robustness to using state-by-year instead of year fixed effects. State-year fixed effects account for unobserved factors that are constant across households within a state-year that jointly determine migration propensity and agricultural outcomes, such as state GDP or state-level agricultural policies. The results are very similar to the baseline estimates.[16]

[15] Figure B2 shows the distribution of household farm size. The vast majority of farms are < 5 acres.

[16] Estimates are also robust to district-year fixed effects, suggesting that the decline in agricultural activity is not driven by district-level shocks such as changes in local economic conditions, infrastructure, etc.

Panel B adds a control for village-level exposure to the National Rural Employment Guarantee Act (NREGA), a large public works program introduced shortly after IHDS Wave I, and which affected local labor and land markets (Imbert and Papp, 2015; Muralidharan et al., 2023). Because household participation in NREGA may reflect its own labor or agricultural conditions, we aggregate the number of household job cards to the village level to construct a less endogenous measure of local program exposure. We set NREGA exposure to zero in Wave I, before it was rolled out. Our estimates remain virtually unchanged from the baseline specification, suggesting that differential exposure to NREGA does not explain our main results.

Panel C tests robustness to including the extensive margin in the sample—households that left or joined agriculture between the two surveys. To do so, we zero-impute outcome variables for landless households. This adds to the sample landless households in Wave I who owned land in Wave II as well as households who owned land in period 1 and then left agriculture in period 2. The main results are robust to this alternative sample.

Panels D-E show that our estimates are robust to alternative shift-share instruments. First, we explore alternative shifts. Panel D redefines z_{idt} to isolate rural-urban migration by restricting the destination choice set (Θ in Equation 1) to urban destinations only, and restricting the sample of out-migrants to rural households only. The results are virtually unchanged, suggesting that our baseline estimates are not driven by any particular migration stream but rather reflect internal migration in general. In Panel E, we use inverse-distance weighted nightlights as the shift[17]. Again, results remain stable, suggesting that our baseline estimates are robust to different measures of the shift.

We next explore alternative measures of the "share". Panel F tests robustness to redefining the migration potential parameter, φ_i, as the number of working age male *or female* household members. Accordingly, M_{idt}^{labor} in the first stage is also redefined as the total number of working-age migrants sent from household i, irrespective of gender. The estimates remain negative and statistically significant, although somewhat larger in magnitude compared to the the baseline specification.[18]

Table A10 tests the sensitivity of our estimates to the choice of the distance decay

[17]Satellite-detected nightlights are considered a strong proxy for local GDP (Henderson et al., 2012). Data are obtained from the DMSP-OLS satellite product at 30 arc second resolution.

[18]This should not be interpreted as evidence that female migration has larger effects than male migration. Rather, redefining the endogenous migration measure and the instrument changes the set of compliers. In the main IV specification, the compliers are households that send an additional working-age male migrant in response to destination income shocks. Here, the compliers are households that send an additional working-age migrant of either gender. Because female migration is considerably less common, these households may represent a more selected group with stronger migration responses to urban opportunities. The larger coefficient therefore reflects a different local average treatment effect.

parameter, τ. While we used $\tau = 1$ to construct the preferred shift-share (Equation 1), columns 1-4 show that the estimates do not change meaningfully when we reconstruct the instrument using distance decay functions ranging from $\tau = 0.5$ to $\tau = 2$.

Lastly, Table A11 reports estimates under alternative choices of clustering for statistical inference. Our baseline estimates report standard errors clustered by district, since households in the same district are exposed to the same destination income shocks. We also test robustness to clustering at the Population Sampling Unit level, a fundamental geographic unit defined by IHDS for the initial stage of sampling, and within which households may share unobserved characteristics. We also cluster by state in case shocks are correlated across districts within the same state. Lastly, we investigate spatial correlation more systematically by estimating Hsiang (2010)'s implementation of Conley (1999) standard errors for kernel cut-off distances ranging from 200km to 500km.[19] Overall, the statistical significance of our estimates of the direct effect of migration on agricultural outcomes is remains largely unchanged across these alternative methods of accounting for spatial and serial correlation.

5.5 Summary of Direct Effects

We have shown that agricultural households in India do not substitute labor with capital when facing migration-induced labor loss. Instead, they reduce agricultural technology adoption and downsize their farms. India's fast urbanization (Figure 1) thus raises a concern about agricultural production and domestic food supply. But the net effect on food supply also depends on general equilibrium adjustments through land and food markets: how do other agricultural households who reside in the same village, or other households in the same state who grow the same crops, react to migrant households' divestment from agriculture? Our detailed household panel data allow us to decompose these partial- and general-equilibrium responses and better reconcile our findings with existing work, which has been limited to more aggregated levels of analysis (e.g., Hornbeck and Naidu (2014)). We turn to this decomposition next and, in doing so, extend the literature and deepen our understanding about the spatial reorganization of agriculture in response to structural change.

[19] We adapt the Hsiang (2010) implementation to an IV setting using the method of Colella et al. (2019).

6 Equilibrium Crop and Land Market Adjustments

The conceptual framework (Section 2) highlights two channels through which migration may generate indirect effects. First, declines in farming activities among migrant-sending households may raise crop prices, encouraging other households to expand production. Second, contractions in cultivated area may reduce land prices, incentivizing farm expansion elsewhere. Both channels imply that agricultural losses among migrant households may be partially offset by gains among non-migrant households. Because migration opportunities are more common near urban centers (Fact 2, Section 3.2), these indirect effects are expected to be strongest in more remote areas, where households are less likely to send migrants but remain exposed to changes in crop and land markets.

We formalize these ideas in a spatial equilibrium model in Appendix C and dedicate the remainder of this section to testing its predictions. Similar to Adao et al. (2019) and Borusyak et al. (2022b), the model shows that the overall effect of migration on agricultural outcomes is composed of additively separable direct and indirect channels. We use this feature to extend our 2SLS setup to incorporate both of these channels.

6.1 Measurement

6.1.1 The Land Market Channel

To measure land market adjustments, we conceive land markets at the village level. The land market adjustment is then measured by aggregate emigration from all other households i' of village j, in district d, excluding household i:

$$M_{ijdt}^{land} = \frac{\sum_{i' \in N_{jdt}/i} Migrants_{i'jdt}}{|N_{jdt}/i|} \tag{5}$$

Intuitively, M_{ijdt}^{land} measures average migration from other households $i' \neq i$ in the land market. $Migrants_{i'jdt}$ is the number of working-age male migrants sent from households i'. N_{jdt}/i is the set of households in village j excluding household i, and $|N_{jdt}/i|$ is the number of elements in N_{jdt}/i. We divide the number of migrants (except household i) by the number of households in village j (except household i) to account for heterogeneous village populations and land endowments. Household i's response to changes in M_{ijdt}^{land} captures adjustments operating through the indirect land channel. In Section 6.5, we show that greater emigration from household i's land market, M_{ijdt}^{land}, reduces our proxy measure for land prices (Table 6).

6.1.2 The Crop Market Channel

To measure crop market adjustments, we describe crop markets as the combination of all households in the state producing similar crops. One concern is that crop choice is endogenous since they depend on crop prices, labor supply, and land supply, which all respond to migration and destination income shocks. We thus characterize crop markets based on (i) similarity in crop-specific suitability across areas and, in a robustness test, (ii) similarity in household's baseline crop portfolios based on IHDS data. As we show below, both measures are plausibly exogenous to destination income shocks in the sample.

To quantify crop similarity, let $\mathbf{x}_d = (x_{d1}, ..., x_{dK})$ denote the vector of agro-climatic suitability scores for the K crops grown in district d, where $x_{dk} \in [0, 1]$ is the suitability of crop k. Households within the same district share the same crop-suitability vector. Crop similarity between household i and i', located in districts $d(i)$ and $d(i')$, is measured as the inverse Euclidean distance between their crop suitability vectors[20]:

$$d(\mathbf{x}_{d(i)}, \mathbf{x}_{d(i')})^{-1} = \left(\sqrt{(x_{d(i)1} - x_{d(i')1})^2 + (x_{d(i)2} - x_{d(i')2})^2 + ... + (x_{d(i)K} - x_{d(i')K})^2} + 1 \right)^{-1}$$

Using this idea of crop similarity, the crop market adjustment is measured as the weighted average emigration from other households i' in the same state s, with weights equal to crop similarity between household i and i' in the baseline period:

$$M_{ijdt}^{crop} = \sum_{i' \in N_s / i} d(\mathbf{x}_{d(i)}, \mathbf{x}_{d(i')})^{-1} Migrants_{i't} \tag{6}$$

We assume state-level crop markets because cross-state trade restrictions in India prevent crop prices from equilibrating across states (Chatterjee, 2023).[21] N_s / i denotes all households in state s excluding household i. Since emigration from other households growing equally suitable crops to i receives more weight in the aggregation, M_{ijdt}^{crop} effectively measures aggregate emigration from the same crop market as household i. Thus, the response of household i to changes in M_{ijdt}^{crop} reflects indirect effects through crop markets. In Section 6.5, we show that household responses to crop market adjustments captures an underlying relationship between M_{ijdt}^{crop} and crop prices (Table 6). Intuitively, if destination income shocks draw agricultural labor away from a rice-growing area, thereby affecting aggregate rice supply and rice prices, then households elsewhere will respond more to

[20] We use d(x)=1/(x+1) to avoid divide-by-zero issues.

[21] We ignore crop trade cost for simplicity. A possible extension could weight emigration of other households by their inverse distance to the focal household. However, since the crop market is heavily regulated within states, we assume the government absorbs inter-state trade costs.

the price change if they also live in a rice-suitable part of the state.

We use gridded crop suitability (1km resolution) from the GAEZ FAO portal (Fischer et al., 2021) to measure x_{dk}. Suitability is an index over the 1980-2010 period, normalized to one. We obtain separate suitability rasters for each major crop in India.[22] District suitability for each crop is computed by extracting means over all cells within a district.

6.2 Estimating Equation

In order to jointly estimate the direct and indirect effects of internal migration, we extend the 2SLS framework in Section 4.3 to include the land and crop market channels:

$$M_{ijdt}^{labor} = \mu_1 z_{ijdt} + \mu_2 s_{dt} + \mu_3 inc_{dt} + \mu_4 M_{ijdt}^{land} + \mu_5 M_{ijdt}^{crop} + \Gamma X'_{dt} + \alpha_i + \gamma_t + \varepsilon_{ijdt} \quad (7)$$

$$Y_{ijdt} = \beta_1 \widehat{M_{ijdt}^{labor}} + \beta_2 M_{ijdt}^{land} + \beta_3 M_{ijdt}^{crop} + \beta_4 s_{dt} + \beta_5 inc_{dt} + \Gamma X'_{dt} + \alpha_i + \gamma_t + \eta_{ijdt} \quad (8)$$

where M_{ijdt}^{land} and M_{ijdt}^{crop} are variables defined above to capture land market and crop market adjustments, respectively. Other terms are the same as Equation 4. The total response of farmers to emigration comprises the direct effect through household migration-induced labor loss, β_1, as well as the indirect effects through land and crop market adjustments, captured by β_2 and β_3, respectively. Only the direct labor channel, M_{ijdt}^{labor}, is instrumented with z_{ijdt}, whereas the market responses are not. This assumes that migration of other households in the village and crop market is unrelated to the decision of household i, conditional on controls and fixed effects.

For land markets, a threat to identification is that households i and i' may respond to village-level shocks (e.g., a local drought or the introduction of an agricultural program) that co-determine village emigration and household agriculture. We therefore control for droughts, temperature, and rainfall in all specifications, as well as for village agricultural extension programs in a robustness check. To the extent that these village shocks manifest as pull factors, these are captured by distance-weighted destination income shocks, s_{dt}. Another threat is that village emigration may affect household agriculture through factor markets other than land. We rule this out by controlling directly for non-land factor market changes in a placebo test (Section 6.4.2).

For crop markets, the concern is that crop-specific shocks, such as disease or trade disruption, can affect both emigration from the crop market and household agricultural outcomes via price channels. We thus control for average district-level crop prices in a

[22] Major crops are defined by the Department of Agriculture, Cooperation & Farmers Welfare as rice, wheat, nutri-cereals, pulses, foodgrains, oilseeds, sugarcane, cotton, and jute & mesta.

robustness check to ensure that identification of β_3 comes from deviations from general crop price trends driven by aggregate supply contractions resulting from emigration from the crop region. We also validate our measure of M_{ijdt}^{crop} by showing that household agriculture does not respond to aggregate emigration from outside the crop market.

A final threat to both the land- and crop-market adjustment estimates is that households in villages or crop regions that later experience greater emigration may already have been on different agricultural trajectories before a given household member migrates. To assess this, we estimate a lead test analogous to Equation 2 with additional terms for M_{ijdt}^{land} and M_{ijdt}^{crop}. We find no relationship between future aggregate emigration and baseline agricultural investment, farm size, or profits (Table A12). The only exception is baseline labor use, which is negatively related to future aggregate emigration. These results provide reassurance that our land and crop market measures are not simply capturing pre-existing trends, but instead reflect market adjustments induced by emigration.

6.3 Results: Direct and Indirect Effects

Table 5 presents estimates of β_1, β_2, and β_3 from Equation 8. Estimates are reported in standard deviation units for ease of comparison. The main result is that direct and indirect effects draw in opposing directions. The direct effect (row 1) remains negative, statistically significant, and similar in magnitude to Table 4. However, column 1 shows that migration-induced declines in agricultural technology are partially offset by non-migrant household responses through land (row 2) and crop (row 3) market adjustments.

When decomposing by technology type, Table A13 columns 1-5 show that technology uptake through crop market adjustments materializes through spending more on seeds, agrochemicals, water, and rental equipment. There is little change in machinery ownership (columns 6-10). We show in Section 6.5 that agricultural expansion through crop market adjustments captures household responses to higher crop prices.

Column 3 of Table 5 shows that downsizing farms through the direct channel is also partially offset by the indirect land market channel (row 2). This corroborates the model prediction that falling land prices induced by the direct effect prompt non-migrant households to expand farm size. We provide evidence for this price mechanism in Section 6.5. Column 4 reports labor responses and indicates that total labor days declines in response to emigration, whereas the indirect effects operate in the opposite direction. We interpret this pattern as reflecting households' increased own-family labor supply in response to higher wage rates in line with the findings of Breza et al. (2021) for lean months.

Lastly, the estimates for farm profits in column 5 show the same directionality through

Table 5: Direct and Indirect Effects of Internal Migration

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants	-1.075***	-0.744***	-1.242***	-1.955***	-0.965***
(direct labor channel)	(0.228)	(0.219)	(0.259)	(0.316)	(0.193)
Village emigration	0.236***	0.189***	0.259***	0.405***	0.162***
(indirect land channel)	(0.053)	(0.048)	(0.055)	(0.077)	(0.044)
Crop region emigration	0.206***	0.030	0.262***	0.445***	0.176***
(indirect crop channel)	(0.057)	(0.057)	(0.057)	(0.084)	(0.041)
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	25924	24966	29342	20906	27458
F-Stat on Direct Effect	64.1	59.4	52.5	56.5	68.8

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Outcomes and all explanatory variables are standardized to have a mean of 0 and a standard deviation of 1. *Male Migrants* is number of working-age male migrants sent from the household, instrumented with inverse-distance population-weighted income interacted with a number of baseline working-age males. *Village emigration* is the leave-one-out average number of working-age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants in the state weighted by inverse Euclidean distance between crop suitability (Section 6.1). Columns 1 and 2 are indices of agricultural expenses and stock of machinery, respectively. Column 3 is farm size, measured as cultivated acres. Labor (column 4) is person-days of labor (household + hired). Farm profit (column 5) is defined as farm income minus expenses. All columns control for origin income, the uninteracted shift, drought, temperature, and rainfall. Standard errors clustered by district.

each channel. Point estimates are precise and economically significant. Whereas households with migrants experience lower farm profits by 0.97σ in response to their own labor loss, agricultural expansion through crop and land market adjustments offset this direct effect. In Section 7.2, we use these coefficients in an accounting exercise to quantify the extent to which aggregate agricultural losses are compensated by market spillovers.

Note that negative direct effects primarily impact households sending many migrants, whereas households sending zero migrants indirectly benefit from agricultural expansion. Since remote households send fewer migrants (Figure 2), we expect a spatial reorganization of agriculture away from peri-urban areas with high emigration toward remote areas with low migration. In Section 6.6, we show that this spatial pattern is indeed borne out in aggregate and high-resolution data, supporting our empirical findings and corroborating our narrative about the spatial reorganization of agriculture.

6.4 Robustness and Placebo Tests

We test the sensitivity of our estimates of crop and land market adjustments with a variety of robustness and placebo tests. The resulting estimates are remarkably stable, and thus our market measures appear robust and well-defined.

6.4.1 Robustness Tests

Tables A14 and A15 test whether correlated village-level shocks drive our estimates of indirect effects. This is important because M_{ijdt}^{land} is uninstrumented in Equation 8, so local shocks may jointly affect village-level emigration incentives and household agricultural decisions. We first control for village-level exposure to NREGA, following the approach described in Section 5.4. We then control directly for environmental shocks (drought, flood, cyclone, or earthquake) and for the presence of agricultural extension programs, both which are recorded in the IHDS village module for all 1400 villages in the sample. Across both exercises, the coefficient on M_{ijdt}^{land} remains highly stable, suggesting that the estimated land market adjustments are not driven by concurrent changes in local labor or land markets, environmental conditions, or agricultural extension programs.

Table A16 shows that estimates of crop market adjustments are stable when controlling for annual district-level prices of households' highest-value crop[23]. This absorbs general crop-level shocks, such as crop disease or trade disruptions, that affect both crop-region emigration and household agriculture via crop prices. As such, β_3 is identified from deviations from general crop price trends specifically driven by supply contractions induced by aggregate emigration from the crop region.

Table A17 tests robustness to an alternative definition of the crop market. For this test, crop similarity weights are defined in crop production space instead of suitability space. Crop-wise production data is available in the 2005 IHDS survey. We define M_{ijdt}^{crop} as in Equation 6, except that $d(\mathbf{x}_i, \mathbf{x}_{i'})^{-1}$ is constructed such that K is the number of possible crops that households report growing, and x_{ik} is crop k's share of actual production for household i. $x_{ik} = 0$ if crop k is not grown. Reassuredly, estimates of β_3 are virtually identical to those in Table 5 when we use this alternative measure of M_{ijdt}^{crop}.

Table A18 reports estimates allowing crop markets to span the whole of India. This allows household i to be affected by the emigration from households growing similarly suitable crops anywhere in India rather than only within the same state. The coefficient on M_{ijdt}^{crop} remains positive across most specifications, and the magnitudes increase. Estimate sensitivity is likely driven by the fact that agricultural markets in India function at the

[23] Estimates are also robust to controlling for linear crop-time trends.

state level, in which case this robustness test may be misspecified. We therefore proceed with our main specification where crop markets are defined at the state level.

Table A19 test robustness to including the extensive margin in the sample. Recall that Table 5 comprises a sample of households that cultivated land in both periods. Households who left or entered agriculture between surveys have missing outcomes when they are landless. We add them to the sample by zero-filling their agricultural technology expenses and machinery. Estimates of the direct and indirect effects remain remarkably stable, suggesting that including the extensive margin of farmer exit and entry in response to labor migration does not change our main conclusions.

6.4.2 Placebo Tests

In addition to the robustness checks, we design placebo tests to validate our measures of crop and land market adjustments and provide added support for the interpretation of our estimates as indirect effects. First, we compute a placebo crop market measure by replacing $d(\mathbf{x}_{d(i)}, \mathbf{x}_{d(i')})^{-1}$ in Equation 6 with $d(\mathbf{x}_{d(i)}, \mathbf{x}_{d(i')})$. Since these weights are not inverted, households growing different crops receive more weight in the aggregation. If our estimates of household responses to crop market adjustments are driven by household i reacting to supply shifts induced by emigration through the crop market, then emigration of households outside the crop market should not impact household i's farm profits. Column 1 of Table A20 controls for this placebo in Equation 8 and shows that its coefficient turns negative and converges to zero. The robust coefficient on M_{ijdt}^{crop} helps confirm that our estimate of β_2 is driven by household i's response to crop market adjustments.

Next, we compute a more sophisticated crop market placebo that accounts for crop substitutability. Even if households i and i' grow different crops, their production may still be linked through prices if they grow substitutes or complements. We, therefore, measure $d(\mathbf{x}_i, \mathbf{x}_{i'})^{-1}$ as the inverse cross-price elasticity between the main crop of household i and i' obtained from Anand et al. (2016)[24]. Taking the inverse implies that households growing unrelated crops receive more weight in the aggregation. Column 2 of Table A20 controls for this placebo and, once again, the placebo coefficient is near-zero. This suggests that households' reaction to crop market adjustments in response to aggregate emigration does not only depend on supply shifts of their main crops within the crop market, but also on supply shifts of substitutes and complements within the market.

[24]Compensated elasticities for India are provided for four relevant crop categories: cereals, pulses, veggies/fruits, and other. We first identify the main crop of each household as the one with the highest share of total production. Second, we categorize each crop into one of the four categories. Lastly, we fill the weight matrix with the absolute value of elasticities for each pair of households.

Table 6: Mechanisms: Impact of Aggregate Emigration on Prices

	(1) Crop Price	(2) Land Price	(3) Land Price	(4) Ag. Wage
Crop region emigration (indirect crop channel)	0.137*** (0.052)			
Village emigration (indirect land channel)		-0.052** (0.025)	-0.055** (0.025)	0.014 (0.025)
Vil. Environmental Shocks	No	No	Yes	Yes
Vil. Agricultural Extension	No	No	Yes	Yes
Aggregation	District-year	Village-year	Village-year	Village-year
District FEs	✓			
Village FEs		✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	586	1222	1100	1086
R^2	0.892	0.667	0.664	0.558

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Coefficients are OLS estimates. Outcome and explanatory variables are standardized to have a mean of 0 and a standard deviation of 1. Crop price is a district-weighted over crops grown in a district, with weights equal to suitability scores. Land price is proxied by annual rental payments. In columns 2-3, regressions are weighted by the number of households renting land. *Crop region emigration* is a district average of the leave-one-out number of migrants *within the state* weighted by crop suitability. *Village emigration* is mean number of working age male migrants in the village. Errors clustered by district in column 1 and by village in columns 2-4.

Lastly, column 3 presents placebo estimates that validate our land market adjustment measure. One concern is that emigration by other agricultural households in the village might also shift non-land factor prices, like wages or input costs, in addition to land prices. We therefore control for emigration by landless households in the village in Equation 8, which affects only non-land indirect channels and therefore disentangles the effect of land and non-land indirect channels on household agriculture. We are well-powered for this test since over half of the households in our sample are non-agricultural. The land market placebo does not affect household farm size, whereas our main land market coefficient remains stable. We conclude that our land market measure, M_{ijdt}^{land}, is indeed picking up farm size adjustments through land markets and not other indirect channels.

6.5 Mechanisms: Land and Crop Markets

We argued throughout the paper that β_2 and β_3 in Equation 8 capture households' indirect responses to land and crop market adjustments, respectively. Our model (Appendix C) shows that prices are the guiding force behind these adjustments. We pro-

vide evidence for the price mechanism using measures of on crop and land prices. Time-varying crop prices are taken from ICRISAT and measured as the weighted average crop price for major crops in each district, with weights equal to the suitability index for each crop (Section 6.1 for data details). Although IHDS does not directly report land rental prices, we proxy for them using annual rental payments reported by households renting in farmland. We average rental payments at the village level and weight regressions by the number of households renting land. This ensures that observations measured with greater precision receive greater weight in determining the estimated coefficients.

We estimate the price effects in response to emigration through the crop and land market using the following regressions:

$$CropPrice_{dt} = \Phi \cdot M_{dt}^{crop} + \alpha_d + \gamma_t + \epsilon_{dt}$$

$$LandPrice_{jt} = \psi \cdot M_{jt}^{land} + \alpha_j + \gamma_t + \epsilon_{jt}$$

where d, j, and t index districts, villages, and time, respectively. $CropPrice$ is the crop price index in district d and $LandPrice$ is our proxy for land rental price in village j. M_{dt}^{crop} and M_{jt}^{land} are measured as before, but aggregated at the district and village level, respectively. The coefficients of interest are Φ and ψ, which are estimated with OLS since M_{jt}^{land} and M_{dt}^{crop} are uninstrumented. If $\Phi > 0$, the crop market response to emigration leads to higher crop prices. Likewise, if $\psi < 0$, then the land market response to emigration reduces land prices. All specifications include location and time fixed effects.

Table 6 presents the estimates of the price effects. Outcomes are reported in standard deviation units. Starting with crop price in column 1, the coefficient is positive, significant and implies that a 1σ increase in emigration from the crop market increases crop prices by 0.14σ. This suggests that crop prices are a key mechanism behind the positive crop market effect in Table 5: declining farm activity through the direct labor channel raises crop prices, leading non-migrant farmers in the crop market to increase production.

Column 2 documents the impact of emigration from the land market on our proxy for land rental prices. The coefficient is negative and significant, implying that emigration from the land market depresses land rents. This supports the price mechanism behind farm expansion through indirect land market adjustments (Table 5): the reduction in cultivation through the direct labor channel increases land availability, thereby reducing land prices and leading to farm expansion among other households in the land market (village) with fewer or no migrants. The land price effect remains stable when controlling for village environmental shocks and agricultural extension programs (column 3).

Finally, column 4 validates our proposed land and crop price mechanism. An alterna-

tive possibility is that migration primarily affects agriculture through rising rural wages, inducing non-migrant households to adjust production in response to higher labor costs rather than changes in land or crop prices. To distinguish between these mechanisms, we estimate the effect of village emigration on agricultural wages (Section 3.1.3). We find no detectable wage response. Note that this does not imply that labor markets are absent. Rather, it suggests that labor market adjustments are not the primary mechanism underlying the indirect effects documented in Section 6.3. This is consistent with our estimates of direct effects, which show that migrant-sending households do not replace migrating family members by hiring additional workers. Instead, they reduce hired labor alongside machinery and other complementary inputs (Table 4). This means that labor demand contracts together with labor supply, limiting upward pressure on equilibrium wages. Taken together, the evidence suggests that indirect responses to migration operate primarily through land and crop market adjustments, rather than labor market adjustments.

6.6 The Spatial Reorganization of Agricultural Land

Thus far, we have uncovered two distinct responses to migration: (i) negative direct effects, in the form of *agricultural contraction* among migrant-sending families, and (ii) positive indirect effects, in the form of *farm expansion* by non-migrant households in shared land and crop markets. In this section we explore what form the land transfers between migrant-sending and non-migrant households take, how these adjustments alter agricultural land use overall, and whether there are any systematic spatial dimensions of those changes which affect the aggregate spatial patterns of land use across India.

When a migrant-sending household scales back cultivation, they can rent out the unused land to another household, give it to sharecroppers, sell the land, convert it to non-agricultural uses, or leave it fallow. The IHDS questionnaire does not provide a complete account of what the land is converted to, so we assemble multiple data sources to create a more comprehensive account land use transitions. IHDS asks households whether they receive cash for renting land, a share of the crop, or both. Table A21 estimates our same IV specification (Equation 4) with these outcomes and finds that migrant-sending households are 6% more likely to participate in sharecropping contracts (column 1), with no detectable change in the likelihood of cash-rental contracts (column 2).

To explore whether any land is left fallow or transitions out of agricultural use altogether, we obtained gridded land cover data from the European Space Agency (ESA) for 2005 and 2012, consolidated the 22 land classes into seven categories based on the IPCC classifications, and then computed the percent of district area under each land category

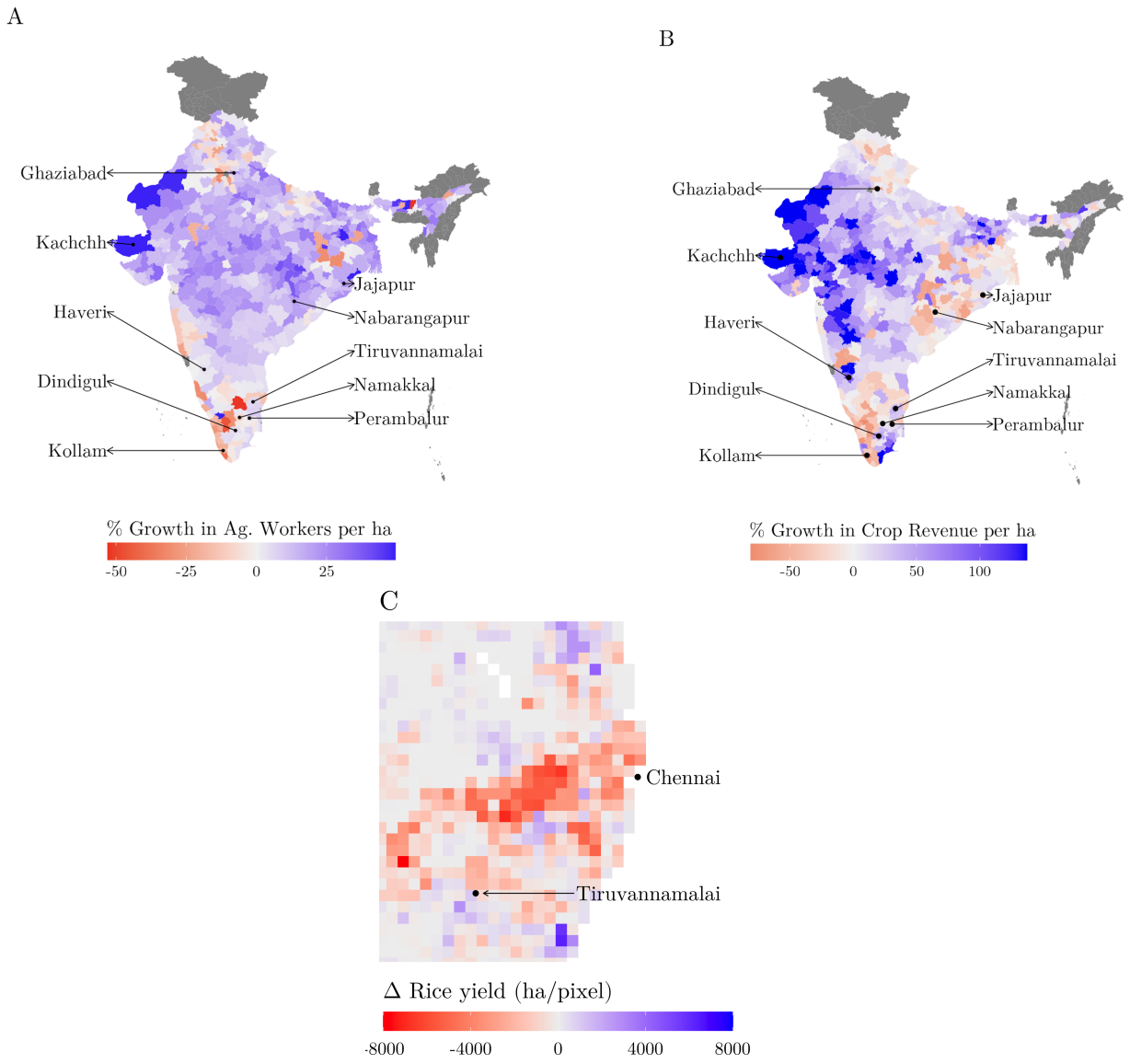


Figure 4: Economic Growth, Labor Exit, and Agricultural Growth (2001-2011)

Note: The time period for each map is 2001-2011. Panel A maps % change in agricultural workers per hectare of cultivated area using data from the 2001 and 2011 Census. Panel B maps the % change in crop revenue per hectare using data from the ICRISAT database. Revenue is measured as value of all crops produced in a district divided by gross cropped area. Values (in both panels) are truncated at the 5th and 95th percentile. Panel C maps the change in harvested rice area per pixel from the GGCP10 satellite product.

in each year. We then aggregated IHDS to district-year means, merged it with the ESA panel, and estimated our 2SLS specification using inverse-distance weighted destination income as an instrument for district-level migration. Unlike the household analysis, the instrument is not interacted with baseline number of household males because the analysis is at the district level. Table A22 shows that labor loss increases forest cover and barren land, consistent with some farmland being abandoned or transitioning out of agriculture.

Our estimates of indirect effects in Table 5 imply that these responses unfold along a spatial gradient: agricultural contraction materializes among peri-urban households, who face low migration costs and send many migrants, whereas agriculture expands among more remote households, for whom migration costs are high and emigration is less common. This implies a spatial reorganization of agriculture that can be tested with geocoded data. We map changes in economic growth, agricultural labor, and production activity across India using population census data and high-resolution satellite imagery. Unlike IHDS, these data report crop revenue per ha as opposed to farm profits.

The maps in Figure 4 validate our empirical findings and help visualize how structural transformation reshapes the geography of agriculture in India. Panels A and B map changes in farm labor and crop revenue per ha from 2001-2011, respectively. The ten fastest-growing cities in the IHDS are labeled, corresponding to the same destination-income variation used to construct our shift-share IV. Two spatial patterns emerge, each consistent with our empirical results. First, near the cluster of high-growth cities in South India, agricultural labor declines in surrounding districts where the migration pull force is strongest (Panel A, red). These same labor-sending districts also experience declining crop revenue (Panel B, red), echoing our estimates of direct effects in Table 4: as males migrate to cities, left-behind households scale back farming. Second, in more remote regions of Central India, where there are no high-growth cities, we observe a compensating *expansion* of agriculture (Panel B, blue). This aligns with our finding that indirect effects are stronger for households further away from the pull of economic growth.[25]

Having documented the spatial reorganization of agriculture across districts, Figure 4C zooms in on even more localized patterns using high-resolution, satellite-based measures of rice yields in the fast-growing regions of South India. Panel C centers on Tiruvannamalai, a city near Chennai and one of the highest-growth cities during this period. Yields decline to the northeast and northwest of the city—areas with significant labor outflows (Panel A). Yet, yields begin to rise beyond these zones as we move into more suburban areas. This pattern mirrors land market spillovers: as migrant-sending households scale back farming, non-migrant neighbors in the land market (village) expand cultivation. In some cases, red and blue pixels even appear side-by-side, vividly capturing this hyper-localized spatial reallocation of agriculture. These visual patterns help rationalize our empirical findings, consistent with the hypothesis that a geographic redistribution of

[25]Table A23 formally explores the spatial correlations evident in Figure 4 by regressing changes in agricultural laborers and crop revenue per hectare between 2001 and 2011 on the district's distance to the nearest of the ten high-growth cities. Column 1 shows that agricultural labor increases with distance, although the coefficient is imprecise. Column 2 reveals a more precise association for crop revenue: districts 100 km farther from growth hubs exhibit roughly 3 percent higher revenue.

Table 7: IV Estimates—District Level Aggregate Impacts of Migration on Agriculture

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants (σ)	1.043**	1.459*	-0.881	-1.737	0.376
	(0.519)	(0.747)	(0.537)	(1.217)	(0.613)
Origin Income (inc_{dt})	Yes	Yes	Yes	Yes	Yes
Outcome SD	-	-	1.941	124.462	14428.409
Explanatory SD	0.160	0.160	0.158	0.161	0.159
District FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	614	618	656	608	632
F-Statistic	4.7	5.1	4.2	4.6	4.9

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Data are aggregated to district means and converted to standard deviations. Regressions are weighted by number of households over which the mean is taken. Male Migrants is the number of working age male migrants sent from the household, instrumented with inverse-distance population weighted income. All columns control for drought, temperature, and rainfall. Columns 1 and 2 are indices for agricultural expenses and stock of machinery, respectively (Section 6.1). Column 3 is farm size, measured as cultivated acres. Labor (column 4) is person-days of labor (household + hired). Farm profits (column 5) is farm revenue net of expenses. Standard errors clustered by district.

agriculture accompanies structural transformation.

7 Discussion

This section contextualizes our findings, first, by connecting to the existing literature and, second, by conducting an accounting exercise to quantify the extent to which direct aggregate agricultural losses are compensated by crop and land market spillovers.

7.1 District-Level Analysis

A key contribution of our paper is the use of household data to disentangle direct and indirect effects of emigration on agriculture. This decomposition is only possible with household data: direct effects unfold within households, whereas indirect effects unfold spatially across households linked through shared markets. Given that the literature studies structural transformation and agriculture at more aggregated geographic scales (see Literature Review), we aggregate our household data and re-estimate the direct effects at these coarser levels to reconcile our findings with previous studies.

Related studies find that labor loss leads to agricultural mechanization using county-level (Hornbeck and Naidu, 2014), district-level (Aggarwal, 2018), or even state-level (Clemens et al., 2018) data. On these broader geographic scales, the effect of labor loss on agriculture reflects both labor-capital substitution *and* broader market adjustments to labor loss. To bridge our findings with this literature, Table 7 presents district-level estimates of Equation 4, where all variables are aggregated to district means. Since market responses are not included in this equation, aggregation to the district level implies that the coefficient entangles both the response of households to emigration of their own household members as well as responses to broader market adjustments within districts from aggregate emigration. District emigration is instrumented with inverse distance-weighted income shocks (s_{dt}) while also controlling for district income (inc_{dt}), in line with Borusyak et al. (2022b). Due to the aggregation, our instrument is not as strong at the district level, but these estimates are intended only to contextualize our findings.

Our aggregate estimates are generally consistent with the main patterns in the literature. Labor migration leads to increased technology adoption at the district level (Table 7, columns 1 and 2), implying that positive indirect effects outweigh negative direct effects in aggregate. Farm profits also increase, although the estimate is statistically insignificant. In contrast, farm size and labor (columns 3 and 4) decline in response to aggregate emigration, although precision is also weak. Overall, these findings suggest that technology adoption by non-migrant households does not fully compensate for direct labor losses in terms of production impacts, as we report in Section 5

The estimates in Table 7, and those in the literature, capture only the subset of broader market responses that manifest within the aggregation region. Yet crop markets can extend beyond districts (Section 6.1), which means that crop market spillovers are insufficiently unaccounted for in these district-level estimates. This may explain why district-level farm profits show a statistically insignificant response despite greater technology adoption: agricultural expansion unfolds through spatial market spillovers in ways that Equation 4 overlooks when aggregated at higher levels.

7.2 How much of the direct agricultural loss does land and crop markets offset?

In aggregate, how much of the crop losses from agricultural divestment among migrant families are compensated by market adjustments? We use our household-level estimates from Equation 8 to calculate below that 90% of the migration-induced decline in farm profits is compensated by domestic market adjustments. These indirect effects are more

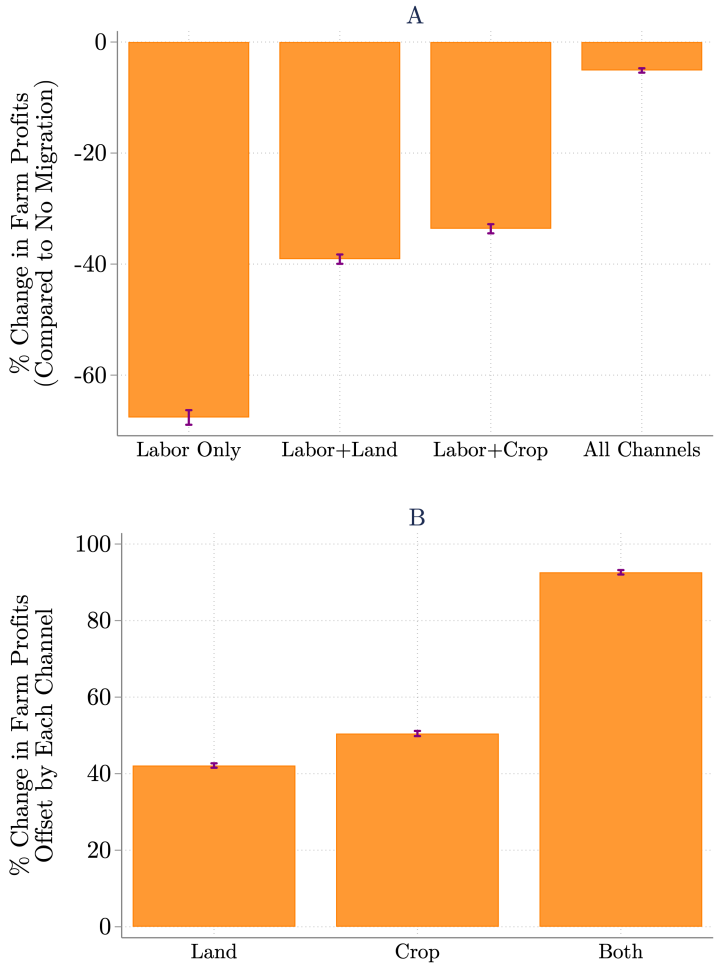


Figure 5: Aggregate Extent of Indirect Effects

Note: Panel A displays the aggregate change in agricultural supply from migration under the four scenarios. *Labor Only* means the crop and land channels are held constant, *Labor + Crop* means the land channel is held constant, and so on. Panel B shows the percent of the *Labor Only* agricultural decline mitigated by the indirect land and crop forces (Equation 9). Confidence intervals computed from 500 bootstrap draws.

pronounced in remote areas, helping explain the spatial reorganization of agriculture away from peri-urban areas and toward more distant regions, as evident in Figure 4.

For each household i in year t, we use the coefficients from Equation 8 to predict household i's farm profits as a function of migration realizations. Let t_1 and t_2 denote IHDS survey wave I and II, respectively. We first define a baseline *No Migration* (NM) scenario in which all migration variables are fixed at t_1:

$$Y_{ijdt_2}^{NM} = \hat{\beta}_1 \widehat{M_{ijdt_1}^{labor}} + \hat{\beta}_2 M_{ijdt_1}^{land} + \hat{\beta}_3 M_{ijdt_1}^{crop} + \alpha_i + \gamma_{t_2}$$

We then define a *Labor Only* (LO) scenario in which both indirect effects are fixed at t_1:

$$Y_{ijdt_2}^{LO} = \hat{\beta_1}\widehat{M_{ijdt_2}^{labor}} + \hat{\beta_2}M_{ijdt_1}^{land} + \hat{\beta_3}M_{ijdt_1}^{crop} + \alpha_i + \gamma_{t_2}$$

In the same way, we define a *Labor and Land* (LL) scenario where the crop market adjustment is fixed at t_1, as well as a *Labor and Crop* (LC) scenario where the land market adjustment is fixed at t_1. We then sum household-level predicted values to construct aggregate counterfactuals: $TotY_{t_2}^{NM}$, $TotY_{t_2}^{LO}$, $TotY_{t_2}^{LL}$ and $TotY_{t_2}^{LC}$, which represent aggregate farm profits without migration, with migration but no market adjustments, with migration plus land market adjustments, and with migration plus crop market adjustments, respectively. Comparisons with in-sample fitted values $TotY_{t_2}$ (the scenario with all channels operational) yield three statistics of interest. The first and second are aggregate values of farm profits with (GE) and without (PE) market adjustments relative to the total farm profits absent migration:

$$PctChange^{GE} = 100 \cdot \left(\frac{TotY_{t_2} - TotY_{t_2}^{NM}}{TotY_{t_2}^{NM}} \right) \quad PctChange^{PE} = 100 \cdot \left(\frac{TotY_{t_2}^{LO} - TotY_{t_2}^{NM}}{TotY_{t_2}^{NM}} \right)$$

The third is the decline in farm profits through the direct channel that is offset by markets as a percent of the counterfactual change absent market adjustments:

$$PctOffset = 100 \cdot \left(\frac{PctChange^{PE} - PctChange^{GE}}{PctChange^{PE}} \right) \tag{9}$$

We follow the same steps to decompose the aggregate effects of the land channel and crop channel separately. Figure 5A shows estimates of the change in aggregate farm profits with and without market spillovers. To account for uncertainty in the underlying point estimates, we repeat the exercise on 500 bootstrapped samples and plot the mean (orange) and confidence intervals (purple error bars). Under the *Labor Only* scenario, with both land and crop market adjustment channels shut off, aggregate migration would have caused a 69% reduction in agricultural profits compared to the *No Migration* counterfactual. This amounts to Rs. 136 million worth of food. When all channels operate, the contraction is more than ten times smaller.

Panel B shows how much of the migration-induced decline in farm profits is mitigated by market adjustments (Equation 9). 41% of agricultural losses through the direct labor channel are mitigated through land markets and 51% through crop markets. Together, the two indirect channels offset 92% of the direct effects of migration. The recovered crops are worth Rs. 125 million, or 45% of the in-sample total crop value in 2012.

It is important to note that these results do not imply a net loss of food across India,

only that we are able to quantify the partial effects of these two observed margins of adjustment. Other unobserved determinants of farm profits, such as climate, are captured by the time fixed effects in our empirical model. The comparison is, therefore, with a hypothetical counterfactual in which all the other determinants of farm profits are considered but migration is reduced to zero.

8 Conclusion

This paper studies how internal migration affects agricultural development and the spatial organization of agriculture. The reallocation of labor from farms to cities is an emblematic feature of economic development. While this process is well-studied, its effects on agriculture in rural areas left behind are relatively unexplored. We track labor migration and agricultural activities with detailed household panel data from India between 2005-2012, a period of rapid economic modernization. We address the endogeneity of migration choice using a shift-share instrument based on distance-weighted destination income shocks interacted with households' potential to benefit from these shocks.

When analyzing structural transformation at an aggregated district level in India, we find evidence consistent with historical US literature that farms mechanize in response to emigration. But our analysis using detailed farm-level data indicate that Indian farmers do not replace labor with capital when facing agricultural labor loss. Instead, they downsize farms and experience lower farm profits. Equilibrium effects on food supply depend on how crop and land markets adjust. We document rising crop prices in response to migration-induced declines in agricultural activity, and falling land prices as more land becomes available. Aggregate effects in the literature thus represent a combination of these direct and indirect general equilibrium effects.

Our analysis uncovers an interesting spatial pattern in the reorganization of agriculture in response to migration. While households in peri-urban areas face agricultural losses, households with no migrants in remote areas benefit from higher crop prices and lower land prices. These remote households expand their farming operations and adopt more technology. In aggregate, our estimates suggest that market spillovers mitigate 90% of the migration-induced food shortage between 2005-2012. The spatial redistribution of agriculture through markets is, therefore, economically significant.

Our results have important distributional implications. We showed that while agricultural development declines in peri-urban areas, it surges in remote areas where low migration rates and poverty are widespread. We thus expect structural transformation to promote income redistribution toward those who do not directly participate in it.

References

Acemoglu, Daron, "When does labor scarcity encourage innovation?," *Journal of Political Economy*, 2010, *118* (6), 1037–1078.

Adao, Rodrigo, Costas Arkolakis, and Federico Esposito, "General equilibrium effects in space: Theory and measurement," Technical Report, National Bureau of Economic Research 2019.

Agarwal, Bina and Malvika Mahesh, "Does the landowner's gender affect self-cultivation and farm productivity? An analysis for India," *The Journal of Development Studies*, 2023, *59* (5), 758–777.

Aggarwal, Shilpa, "Do rural roads create pathways out of poverty? Evidence from India," *Journal of Development Economics*, 2018, *133*, 375–395.

Allen, Treb and Costas Arkolakis, "Economic Activity across Space," *The Journal of Economic Perspectives*, 2023, *37* (2), 3–28.

Alvarez-Cuadrado, Francisco and Markus Poschke, "Structural change out of agriculture: Labor push versus labor pull," *American Economic Journal: Macroeconomics*, 2011, *3* (3), 127–58.

Anand, Rahul, Naresh Kumar, and Mr Volodymyr Tulin, "Understanding India's food inflation: The role of demand and supply factors," *IMF Working Paper No. 16/2*, 2016.

Anderson, Michael L, "Multiple inference and gender differences in the effects of early intervention: A reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects," *Journal of the American statistical Association*, 2008, *103* (484), 1481–1495.

Asher, Sam, Alison Campion, Douglas Gollin, and Paul Novosad, "The Long-run Development Impacts of Agricultural Productivity Gains: Evidence from Irrigation Canals in India," *Mimeo*, 2023.

Blakeslee, David, Aaditya Dar, Ram Fishman, Samreen Malik, Heitor S Pellegrina, and Karan Singh Bagavathinathan, "Irrigation and the spatial pattern of local economic development in India," *Journal of Development Economics*, 2023, *161*, 102997.

Borusyak, Kirill and Peter Hull, "Nonrandom exposure to exogenous shocks," *Econometrica*, 2023, *91* (6), 2155–2185.

_ , _ , **and Xavier Jaravel**, "Quasi-experimental shift-share research designs," *The Review of Economic Studies*, 2022, *89* (1), 181–213.

_ , _ , **and** _ , "A practical guide to shift-share instruments," *Journal of Economic Perspectives*, 2025, *39* (1), 181–204.

_ , **Rafael Dix-Carneiro, and Brian Kovak**, "Understanding Migration Responses to Local Shocks," *Available at SSRN 4086847*, 2022.

Brewer, Julia, Ashley Larsen, and Frederik Noack, "The land use consequences of rural to urban migration," *American Journal of Agricultural Economics*, 2022.

Breza, Emily, Supreet Kaur, and Yogita Shamdasani, "Labor rationing," *American Economic Review*, 2021, *111* (10), 3184–3224.

Bryan, Gharad and Melanie Morten, "The aggregate productivity effects of internal migration: Evidence from Indonesia," *Journal of Political Economy*, 2019, *127* (5), 2229–2268.

Bustos, Paula, Bruno Caprettini, and Jacopo Ponticelli, "Agricultural productivity and structural transformation: Evidence from Brazil," *American Economic Review*, 2016, *106* (6), 1320–65.

Caprettini, Bruno and Hans-Joachim Voth, "Rage against the machines: Labor-saving technology and unrest in industrializing England," *American Economic Review: Insights*, 2020, *2* (3), 305–20.

Caunedo, Julieta and Namrata Kala, "Mechanizing Agriculture," Working Paper 29061, National Bureau of Economic Research July 2021.

Chatterjee, Shoumitro, "Market Power and Spatial Competition in Rural India," *The Quarterly Journal of Economics*, 2023.

Clemens, Michael A, Ethan G Lewis, and Hannah M Postel, "Immigration restrictions as active labor market policy: Evidence from the mexican bracero exclusion," *American Economic Review*, 2018, *108* (6), 1468–87.

Colella, Fabrizio, Rafael Lalive, Seyhun Orcan Sakalli, and Mathias Thoenig, "Inference with arbitrary clustering," *IZA Discussion Paper No. 12584*, 2019.

Conley, Timothy G, "GMM estimation with cross sectional dependence," *Journal of econometrics*, 1999, *92* (1), 1–45.

Dinkelman, Taryn and Martine Mariotti, "The long-run effects of labor migration on human capital formation in communities of origin," *American Economic Journal: Applied Economics*, 2016, *8* (4), 1–35.

_ , Grace Kumchulesi, and Martine Mariotti, "Labor migration, capital accumulation, and the structure of rural labor markets," *Review of Economics and Statistics*, 2024, pp. 1–46.

Duarte, Margarida and Diego Restuccia, "The role of the structural transformation in aggregate productivity," *The quarterly journal of economics*, 2010, *125* (1), 129–173.

Emerick, Kyle, "Agricultural productivity and the sectoral reallocation of labor in rural India," *Journal of Development Economics*, 2018, *135*, 488–503.

Fink, Günther, B Kelsey Jack, and Felix Masiye, "Seasonal liquidity, rural labor markets, and agricultural production," *American Economic Review*, 2020, *110* (11), 3351–3392.

Fischer, Gunther, FO Nachtergaele, HT Van Velthuizen, F Chiozza, G Franceschini, M Henry, D Muchoney, and S Tramberend, *Global Agro-Ecological Zones v4–Model documentation*, Food & Agriculture Org., 2021.

Foster, Andrew D and Mark R Rosenzweig, "Economic development and the decline of agricultural employment," *Handbook of development economics*, 2007, *4*, 3051–3083.

_ and _ , "Are There Too Many Farms in the World? Labor Market Transaction Costs, Machine Capacities, and Optimal Farm Size," *Journal of Political Economy*, 2022, *130* (3), 636–680.

Gao, Xuwen, Wenquan Liang, Ahmed Mushfiq Mobarak, and Ran Song, "Restrictions on Migration Create Gender Inequality: The Story of China's Left-Behind Children," *NBER Working Paper No. 30990*, 2022.

Garg, Teevrat, Maulik Jagnani, and Hemant K Pullabhotla, "Rural roads, farm labor exits, and crop fires," *American Economic Journal: Economic Policy*, 2024, *16* (3), 420–450.

Gibson, John, David McKenzie, Halahingano Rohorua, and Steven Stillman, "The long-term impacts of international migration: Evidence from a lottery," *The World Bank Economic Review*, 2018, *32* (1), 127–147.

Gollin, Douglas, David Lagakos, and Michael E. Waugh, "THE AGRICULTURAL PRODUCTIVITY GAP," *The Quarterly Journal of Economics*, 2014, *129* (2), 939–994.

Harris, John R and Michael P Todaro, "Migration, unemployment and development: a two-sector analysis," *The American economic review*, 1970, pp. 126–142.

Henderson, J Vernon, Adam Storeygard, and David N Weil, "Measuring economic growth from outer space," *American economic review*, 2012, *102* (2), 994–1028.

Hoffmann, Lars, Gebhard Günther, Dan Li, Olaf Stein, Xue Wu, Sabine Griessbach, Yi Heng, Paul Konopka, Rolf Müller, Bärbel Vogel et al., "From ERA-Interim to ERA5: the considerable impact of ECMWF's next-generation reanalysis on Lagrangian transport simulations," *Atmospheric Chemistry and Physics*, 2019, *19* (5), 3097–3124.

Hornbeck, Richard and Suresh Naidu, "When the levee breaks: black migration and economic development in the American South," *American Economic Review*, 2014, *104* (3), 963–90.

Hossain, Marup and Ahmed Mushfiq Mobarak, "When Does Emigration Reduce Domestic Economic Activities? Evidence on the Loss of Managerial Capital from a Visa Lottery," *Unpublished manuscript*, 2026.

Hsiang, Solomon M, "Temperatures and cyclones strongly associated with economic production in the Caribbean and Central America," *Proceedings of the National Academy of sciences*, 2010, *107* (35), 15367–15372.

Huber, Kilian, "Estimating general equilibrium spillovers of large-scale shocks," *The Review of Financial Studies*, 2023, *36* (4), 1548–1584.

Imbert, Clement and John Papp, "Labor market effects of social programs: Evidence from india's employment guarantee," *American Economic Journal: Applied Economics*, 2015, *7* (2), 233–263.

Janvry, Alain De, Kyle Emerick, Marco Gonzalez-Navarro, and Elisabeth Sadoulet, "Delinking land rights from land use: Certification and migration in Mexico," *American Economic Review*, 2015, *105* (10), 3125–3149.

Kaboski, Joseph P, Jianyu Lu, Wei Qian, and Lixia Ren, "Melitz Meets Lewis: The Impacts of Roads on Structural Transformation and Businesses," *NBER WP No. 32448*, 2024.

Karlan, Dean, Robert Osei, Isaac Osei-Akoto, and Christopher Udry, "Agricultural decisions after relaxing credit and risk constraints," *The Quarterly Journal of Economics*, 2014, *129* (2), 597–652.

Khanna, Gaurav, Emir Murathanoglu, Caroline Theoharides, and Dean Yang, "Abundance from abroad: Migrant income and long-run economic development," *American Economic Review*, 2026, *116* (4), 1540–1577.

Kinnan, Cynthia, Shing-Yi Wang, and Yongxiang Wang, "Access to migration for rural households," *American Economic Journal: Applied Economics*, 2018, *10* (4), 79–119.

Kleibergen, Frank and Richard Paap, "Generalized reduced rank tests using the singular value decomposition," *Journal of econometrics*, 2006, *133* (1), 97–126.

Lagakos, David and Michael E Waugh, "Selection, agriculture, and cross-country productivity differences," *American Economic Review*, 2013, *103* (2), 948–80.

Lewis, William Arthur, "Economic development with unlimited supplies of labour," 1954.

Manuelli, Rodolfo E and Ananth Seshadri, "Frictionless technology diffusion: The case of tractors," *American Economic Review*, 2014, *104* (4), 1368–91.

Mendola, Mariapia, "Migration and technological change in rural households: Complements or substitutes?," *Journal of Development Economics*, 2008, *85* (1-2), 150–175.

Mobarak, Ahmed Mushfiq, Iffath Sharif, and Maheshwor Shrestha, "Returns to international migration: Evidence from a Bangladesh-Malaysia visa lottery," *American Economic Journal: Applied Economics*, 2023, *15* (4), 353–388.

Morten, Melanie and Jaqueline Oliveira, "The effects of roads on trade and migration: Evidence from a planned capital city," *NBER Working Paper*, 2018, *22158*, 1–64.

Moscona, Jacob, "Agricultural development and structural change within and across countries," Technical Report, Mimeo, MIT 2019.

Muralidharan, Karthik, Paul Niehaus, and Sandip Sukhtankar, "General equilibrium effects of (improving) public employment programs: Experimental evidence from India," *Econometrica*, 2023, *91* (4), 1261–1295.

Pellegrina, Heitor S and Sebastian Sotelo, "Migration, Specialization, and Trade: Evidence from Brazil's March to the West," Technical Report, National Bureau of Economic Research 2021.

Restuccia, Diego, Dennis Tao Yang, and Xiaodong Zhu, "Agriculture and aggregate productivity: A quantitative cross-country analysis," *Journal of monetary economics*, 2008, *55* (2), 234–250.

Rozelle, Scott, J Edward Taylor, and Alan DeBrauw, "Migration, remittances, and agricultural productivity in China," *American Economic Review*, 1999, *89* (2), 287–291.

Schwartz, Aba, "Interpreting the effect of distance on migration," *Journal of political economy*, 1973, *81* (5), 1153–1169.

Tian, Yuan, Junjie Xia, and Rudai Yang, "Trade-induced urbanization and the making of modern agriculture," *Working Paper*, 2023.

Veljanoska, Stefanija, "Do remittances promote fertilizer use? The case of Ugandan farmers," *American Journal of Agricultural Economics*, 2022, *104* (1), 273–293.

Vollrath, Dietrich, "How important are dual economy effects for aggregate productivity?," *Journal of development economics*, 2009, *88* (2), 325–334.

Appendix - For Online Publication Only

A Appendix Tables

Table A1: Migrant Distribution by Type by Age Cohort

	Male Migrated for School	Male Migrated for Work
0-4	0.29	0.21
5-9	0.55	0.03
10-14	0.54	0.06
15-19	0.41	0.31
20-24	0.24	0.55
25-29	0.06	0.75
30-34	0.01	0.80
35-39	0.01	0.83
40-44	0.00	0.83
45-49	0.00	0.83
50-54	0.00	0.81
55-59	0.00	0.80
60+	0.00	0.57
Total	0.18	0.58

Note: Data from IHDS Wave I (2004-05). Each row denotes an age cohort. Values denote the share of migrants in each age group belonging to each migrant type.

Table A2: Summary Statistics: Migrant Profiles

	# Migrants	Share	SD
A: Gender			
Male	13994	0.79	0.41
Female	3656	0.21	0.41
B: Status			
Student	4755	0.27	0.44
Working	11928	0.68	0.47
Neither	967	0.05	0.23
C: Destination			
Within State	11426	0.65	0.48
Out of State	6224	0.35	0.48
D: Stream			
Rural-Rural	5395	0.31	0.46
Rural-Urban	7898	0.45	0.50
Urban-Rural	2061	0.12	0.32
Urban-Urban	1182	0.07	0.25

Note: The table describes data on migrants in each period. There are 4,377 migrants in Wave I and 13,273 in wave II. Subgroups are mutually exclusive. "Rural-Rural" indicates the origin and destination were rural, "Rural-Urban" indicates the origin is rural and the destination is urban, and so on.

Table A3: Labor-to-Land Ratio by Farm Size

	Small (0-2 ac.)		Medium (2-4 ac.)		Large (4+ ac.)	
	Mean	SD	Mean	SD	Mean	SD
Family person-days/acre	335.22	777.37	128.43	421.86	70.45	119.03
Hired person-days/acre	14.83	46.98	12.82	112.23	8.92	23.98
Hired share of total labor	0.08	0.15	0.11	0.16	0.14	0.17

Note: Data are a panel for all land-owning households. Farms are placed into size bins based on baseline farm size. Family and hired person-days of labor refer to the past year.

Table A4: First Stage Results

	(1)	(2)	(3)
z_{idt}: Wt. Income × Working Age Males	0.310***	0.353***	0.356***
	(0.050)	(0.046)	(0.046)
Wt. Income (s_{dt})	No	Yes	Yes
Origin Income (inc_{dt})	No	No	Yes
HH FEs	✓	✓	✓
Year FEs	✓	✓	✓
KP (2006) F-Stat	37.72	58.94	59.37
Observations	27462	27462	27462

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. The outcome is number of working age male migrants sent from household i. "Wt. Income" is inverse-distance, population-weighted income (s_{dt}). "Working Age Males" is the number of working-age male household residents (φ_i). "Origin Income" is mean per capita district income, inc_{dt}. The units of z_{idt} are standard deviations. All specifications control for drought, temperature, and rain. Standard errors clustered by district.

Table A5: OLS Estimates—Direct Effect of Migration on Agricultural Development

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants (σ)	0.022***	0.002	0.018***	0.003	0.002
	(0.007)	(0.007)	(0.006)	(0.011)	(0.007)
Weather Controls	Yes	Yes	Yes	Yes	Yes
Outcome SD	1.017	1.028	3.621	211.329	26744.428
Explanatory SD	0.524	0.521	0.533	0.518	0.525
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	25928	24970	29346	20910	27462
R^2	0.742	0.739	0.771	0.645	0.697

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Outcomes and explanatory variables are in standardized units. Male Migrants is the number of working age male migrants in the household. Columns 1 and 2 are indices for agricultural expenses and stock of machinery, respectively (Section 6.1). Column 3 is farm size, measured by cultivated acres. Labor (column 4) is person-days of labor (household + hired workers). Farm profits (column 5) is farm income net of expenses. All specifications control for drought, temperature, and rainfall. Standard errors clustered by district.

Table A6: Direct Effect of Migration of Labor-Capital Ratio

	OLS	IV
	(1)	(2)
	Log(K/L)	Log(K/L)
Male Migrants (σ)	0.030**	0.067
	(0.013)	(0.177)
Wt. Income (s_{dt})	No	Yes
Origin Income (inc_{dt})	No	Yes
HH FEs	✓	✓
Year FEs	✓	✓
Observations	19092	19092
F-Statistic		47.1

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. The outcome is total agricultural expenses (K) divided by person-days of labor (L). We transform this by log(x+0.01) to avoid divide-by-zero errors. Male Migrants is the number of working age male migrants in the household. Column 1 is the OLS estimate. In column 2, migration is instrumented with inverse-distance population weighted income interacted with the number of baseline working age males in the household. All specifications control for drought, temperature, and rainfall. Standard errors clustered by district.

Table A7: IV Estimates—Direct Effect of Migration on Technology Adoption

	Agricultural Expenses (Rs.)					Machinery (Num. Owned)				
	(1) Seeds	(2) Fertilizer	(3) Pesticide	(4) Water	(5) Rentals	(6) Tubewell	(7) Pumps	(8) Bullcart	(9) Tractor	(10) Thresher
Male Migrants (σ)	-1.048***	-1.457***	-0.762***	-0.580**	-0.801***	-0.412***	-0.469*	-0.976***	-0.362	-0.606***
	(0.229)	(0.277)	(0.190)	(0.253)	(0.189)	(0.152)	(0.248)	(0.218)	(0.245)	(0.209)
Wt. Income (s_{dt})	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Origin Income (inc_{dt})	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Outcome SD	2076.457	2802.867	1131.968	709.926	1583.843	0.409	0.523	0.382	0.257	0.220
Explanatory SD	0.524	0.523	0.519	0.521	0.524	0.521	0.520	0.519	0.520	0.519
HH FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	25852	25454	24054	23496	24602	24692	23862	24042	23882	23746
F-Statistic	54.4	50.6	47.0	45.8	52.7	53.6	56.3	53.1	52.8	53.6

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Outcomes and explanatory variables are in standardized units. Male Migrants is the number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with the number of baseline working age males in the household. Columns 1-5 are measured in Rs. spent in the past year. Water refers to purchases of irrigation water. Rentals include both hired equipment and animals. Column 6-10 are measured in quantities. Pumps include both electric and diesel pumps. All specification control for district drought, temperature, and rainfall. Standard errors clustered by district.

Table A8: IV Estimates—Recentering the Shift-Share Instrument

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants (σ)	-1.099***	-0.745***	-1.302***	-1.930***	-1.010***
	(0.254)	(0.228)	(0.299)	(0.354)	(0.215)
Recentered IV	Yes	Yes	Yes	Yes	Yes
Wt. Income	Yes	Yes	Yes	Yes	Yes
Origin Income (inc_{dt})	Yes	Yes	Yes	Yes	Yes
Outcome SD	-	-	3.621	211.329	26744.428
Explanatory SD	0.524	0.521	0.533	0.518	0.525
HH FEs	✓	✓	✓	✓	✓
District × Year FEs	✓	✓	✓	✓	✓
Observations	25928	24970	29346	20910	27462
F-Statistic	46.5	47.9	40.3	44.6	52.3

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Outcomes and explanatory variables are in standardized units. Male Migrants is the number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with the number of baseline working age males in the household. "Recentered IV" is computed following Borusyak and Hull (2023). "Wt. Income" is inverse-distance, population-weighted income. "Origin Income" is district-level mean per-capita income. Columns 1 and 2 are indices for agricultural expenses and machinery, respectively (Section 6.1). Column 3 is farm size, measured as cultivated acres. Column 4 is labor, measured as total person-days of labor (family + hired workers). Profit (column 5) is farm income minus expenses. All specifications control for drought, temperature, and rainfall. Standard errors clustered by district.

Table A9: Robustness Checks: Direct Effects (IV Estimates)

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Panel A: State-Year FEs					
Male Migrants (σ)	-0.820***	-0.578***	-1.176***	-1.756***	-1.044***
	(0.181)	(0.174)	(0.221)	(0.276)	(0.189)
Observations	25928	24970	29346	20910	27462
Panel B: Controlling for NREGA					
Male Migrants (σ)	-1.084***	-0.743***	-1.236***	-1.918***	-0.977***
	(0.238)	(0.220)	(0.266)	(0.333)	(0.200)
Observations	25928	24970	29346	20910	27462
Panel C: Extensive Margin					
Male Migrants (σ)	-1.336***	-0.972***	-1.207***	-1.980***	-1.117***
	(0.267)	(0.237)	(0.219)	(0.300)	(0.210)
Observations	69756	68698	75334	63970	72760
Panel D: Rural-Urban Migration					
Male Migrants (σ)	-1.125***	-0.755***	-1.293***	-2.065***	-0.974***
	(0.256)	(0.249)	(0.280)	(0.355)	(0.209)
Observations	25148	24288	28176	20256	26536
Panel E: Nightlights Shift-Share					
Male Migrants (σ)	-1.166***	-0.695***	-1.397***	-2.208***	-1.183***
	(0.276)	(0.246)	(0.309)	(0.408)	(0.236)
Observations	25928	24970	29346	20910	27462
Panel F: All Gender Migration					
Migrants (σ)	-2.093***	-1.546***	-2.566***	-3.451***	-1.814***
	(0.595)	(0.509)	(0.702)	(0.849)	(0.441)
Observations	25928	24970	29346	20910	27462

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Outcomes and explanatory variables are in standardized units. Male Migrants is the number of working age male migrants in the household. See Table 4 footnote for details on the migration IV). Panel A includes state-year FEs and remaining panels only include year FEs. Panel B controls for total number of NREGA job cards held in the village-year. Panel C adds households who left and joined agriculture between survey waves to the estimation sample. Panel D restricts to rural-urban migration only. Panel E uses distance-weighted nightlights as the shift. In Panel F, the explanatory variable and the IV "share" includes both males and females. All specifications control for drought, temperature, and rain. Standard errors clustered by district.

Table A10: Robustness—Alternative Distance Matrices

Outcome variable = farm profits	(1) τ=0.5	(2) τ=1	(3) τ=1.5	(4) τ=2
Male Migrants (σ)	-0.976***	-0.978***	-0.972***	-0.970***
	(0.194)	(0.199)	(0.221)	(0.290)
Wt. Income	Yes	Yes	Yes	Yes
Origin Income (inc_{dt})	Yes	Yes	Yes	Yes
Outcome SD	1.042	1.042	1.042	1.042
Explanatory SD	0.525	0.525	0.525	0.525
HH FEs	✓	✓	✓	✓
District × Year FEs	✓	✓	✓	✓
Observations	27462	27462	27462	27462
F-Statistic	59.5	59.4	47.3	21.1

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Outcomes and explanatory variables are in standardized units. The outcome in all columns is farm profits. τ is the distance decay parameter used to construct inverse-distance weighted incomes. Male Migrants is the number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with the number of baseline working age males in the household. "Wt. Income" is inverse-distance, population-weighted income. "Origin Income" is district-level mean per-capita income. All specifications control for drought, temperature, and rainfall. Standard errors clustered by district.

Table A11: Robustness: Alternative Standard Error Clustering

	Technology Expenses	Machinery Ownership	Cultivated Acres	Labor Days	Farm Profit
Coefficient on Migration	-1.081426	-.7373017	-1.23806	-1.912697	-.9780268
SE: District	.2393644	.2221432	.2664669	.3330062	.1991919
SE: PSU	.216698	.1905592	.2395342	.326024	.1845668
SE: State	.3148779	.2506073	.3431825	.4170227	.279894
SE: Conley (200km radius)	.243686	.16859	.2034503	.3146775	.2038013
SE: Conley (500km radius)	.1838671	.1093158	.2158298	.2900143	.1907997

Note: The first row replicates the main coefficient estimates from Table 4. Conley standard errors are estimated in a 2SLS setting using the Colella et al. (2019) implementation.

Table A12: Lead Test: Direct and Indirect Channels

Outcomes are for 2005	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Distance Wt. Income (t=2012)	0.074	0.060	-0.082*	0.063	-0.022
	(0.086)	(0.049)	(0.047)	(0.067)	(0.067)
Village emigration (t=2012)	-0.029	-0.035*	0.016	0.024*	-0.026
	(0.019)	(0.021)	(0.024)	(0.014)	(0.017)
Crop region emigration (t=2012)	0.689	0.336	0.017	0.608***	-0.051
	(0.481)	(0.349)	(0.293)	(0.188)	(0.223)
Household Males	Yes	Yes	Yes	Yes	Yes
Origin Income	Yes	Yes	Yes	Yes	Yes
State FEs	✓	✓	✓	✓	✓
Observations	16872	15810	18211	15206	18820
R^2	0.279	0.211	0.204	0.249	0.143

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Data are a cross section of households. Outcomes are for 2005 and are in standardized units. *Village emigration* is the leave-one-out average number of working age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants in the state weighted by crop suitability (see section 6.1 for measurement). Column 1 and 2 are indices of agricultural technology expenses and machinery ownership. Column 3 is farm size, measured as cultivated acres. Column 4 person-days of labor (hired + family). Farm profits (Column 5) is farm income net of expenses. All regressions include state fixed effects and control for district income and number of household working-age males in 2005. Standard errors clustered by state.

Table A13: Second Stage—Direct and Indirect Effects of Migration on Agricultural Development

	Agricultural Expenses (Rs.)					Machinery (Num. Owned)				
	(1) Seeds	(2) Fertilizer	(3) Pesticide	(4) Water	(5) Rentals	(6) Tubewell	(7) Pumps	(8) Bullcart	(9) Tractor	(10) Thresher
Male Migrants (direct labour channel)	-1.040*** (0.216)	-1.448*** (0.259)	-0.762*** (0.183)	-0.584** (0.254)	-0.802*** (0.186)	-0.415*** (0.151)	-0.480* (0.253)	-0.990*** (0.221)	-0.373 (0.249)	-0.617*** (0.210)
Village emigration (indirect land channel)	0.224*** (0.048)	0.311*** (0.059)	0.147*** (0.044)	0.140*** (0.054)	0.197*** (0.045)	0.131*** (0.040)	0.112** (0.053)	0.237*** (0.052)	0.085 (0.057)	0.123** (0.053)
Crop region emigration (indirect crop channel)	0.204*** (0.052)	0.266*** (0.064)	0.204*** (0.051)	0.117** (0.057)	0.122** (0.052)	-0.078 (0.055)	0.045 (0.062)	0.048 (0.054)	0.085* (0.043)	0.025 (0.044)
Wt. Income	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Origin Income	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HH FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	25848	25450	24050	23492	24598	24690	23860	24040	23880	23744
F-Stat on Direct Effect	63.8	58.7	53.4	51.2	60.1	57.8	58.4	55.9	54.7	55.9

Note: $^{*}\ p < .1$, $^{**}\ p < .05$, $^{***}\ p < .01$. All variables are in standardized units. *Male Migrants* is number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with number of baseline working age males. *Village emigration* is the leave-one-out average number of working age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants in the state weighted by crop suitability (see section 6.1 for measurement). Columns 1-5 are measured in Rs. spent in the past year. Water refers to purchases of irrigation water. Rentals include both hired equipment and animals. Column 6-10 are measured in quantities. Pumps include both electric and diesel pumps. All columns control for origin income, the uninteracted shift, drought, temperature, and rain. Standard errors clustered by district.

Table A14: Robustness of GE Estimates: Controlling for NREGA

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants (direct labour channel)	-1.077***	-0.749***	-1.238***	-1.959***	-0.964***
	(0.227)	(0.217)	(0.257)	(0.315)	(0.193)
Village emigration (indirect land channel)	0.233***	0.186***	0.261***	0.403***	0.164***
	(0.052)	(0.048)	(0.055)	(0.076)	(0.044)
Crop region emigration (indirect crop channel)	0.208***	0.032	0.260***	0.448***	0.175***
	(0.057)	(0.056)	(0.057)	(0.084)	(0.041)
NREGA Exposure	Yes	Yes	Yes	Yes	Yes
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	25924	24966	29342	20906	27458
F-Stat on Direct Effect	64.3	59.4	52.5	56.5	69.1

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. All variables are in standardized units. *Male Migrants* is number of working-age male migrants sent from the household, instrumented with inverse-distance population-weighted income interacted with a number of baseline working-age males. *Village emigration* is the leave-one-out average number of working-age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants in the state weighted by inverse Euclidean distance between crop suitability (Section 6.1). NREGA exposure total job cards held in a village-year. Columns 1 and 2 are indices of agricultural expenses and stock of machinery, respectively. Column 3 is farm size, measured as cultivated acres. Labor (column 4) is person-days of labor (household + hired). Farm profit (column 5) is farm income net of expenses. All columns control for origin income, the uninteracted shift, drought, temperature, and rain. Standard errors clustered by district.

Table A15: Robustness of GE Estimates: Village Level Shocks

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants (direct labour channel)	-1.115*** (0.238)	-0.720*** (0.231)	-1.292*** (0.272)	-2.058*** (0.340)	-0.981*** (0.203)
Village emigration (indirect land channel)	0.252*** (0.057)	0.188*** (0.051)	0.269*** (0.059)	0.449*** (0.083)	0.170*** (0.047)
Crop region emigration (indirect crop channel)	0.210*** (0.062)	0.027 (0.058)	0.251*** (0.060)	0.461*** (0.091)	0.173*** (0.044)
Vil. Enviro Shocks	Yes	Yes	Yes	Yes	Yes
Vil. Ag. Extension	Yes	Yes	Yes	Yes	Yes
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	24192	23356	27090	19384	25518
F-Stat on Direct Effect	63.1	58.2	51.5	54.3	66.4

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. All variables are in standardized units. *Male Migrants* is number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with number of baseline working age males. *Village emigration* is the leave-one-out average number of working age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants across households in the state weighted by inverse euclidean distance between actual crop portfolios between household i and i'. Columns 1 and 2 are indices for agricultural expenses and stock of machinery, respectively. Column 3 is farm size, measured as cultivated acres. Labor (column 4) is person-days of labor (household + hired workers). Farm profit (column 5) is farm income net of expenses. Standard errors clustered by district. All columns control for origin income, the uninteracted shift, a village-level environmental shock dummy, and a village-level dummy for agricultural extension programmes. Standard errors clustered by district.

Table A16: Robustness of GE Estimates: Crop Prices

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants (direct labour channel)	-1.062*** (0.232)	-0.751*** (0.224)	-1.282*** (0.283)	-1.897*** (0.321)	-0.998*** (0.201)
Village emigration (indirect land channel)	0.237*** (0.054)	0.189*** (0.048)	0.275*** (0.060)	0.402*** (0.078)	0.171*** (0.046)
Crop region emigration (indirect crop channel)	0.211*** (0.058)	0.026 (0.057)	0.277*** (0.061)	0.437*** (0.083)	0.197*** (0.043)
Crop Price	Yes	Yes	Yes	Yes	Yes
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	25054	24044	26462	20090	25194
F-Stat on Direct Effect	59.7	57.2	47.0	51.2	61.3

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. All variables are in standardized units. *Male Migrants* is number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with number of baseline working age males. *Village emigration* is the leave-one-out average number of working age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants across households in the state weighted by inverse euclidean distance between actual crop portfolios between household i and i'. Columns 1 and 2 are indices for agricultural expenses and stock of machinery, respectively. Column 3 is farm size, measured as cultivated acres. Labor (column 4) is person-days of labor (household + hired workers). Farm profit (column 5) is farm income net of expenses. Standard errors clustered by district. All columns control for origin income, the uninteracted shift, crop prices, drought, temperature, and rain. Standard errors clustered by district.

Table A17: Robustness of GE Estimates: Crop Markets in Crop Production Space

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants	-1.166***	-0.829***	-1.513***	-2.451***	-1.175***
(direct labour channel)	(0.302)	(0.294)	(0.380)	(0.464)	(0.262)
Village emigration	0.262***	0.225***	0.328***	0.536***	0.215***
(indirect land channel)	(0.070)	(0.066)	(0.082)	(0.119)	(0.061)
Crop region	0.300***	0.057	0.302***	0.596***	0.221***
emigration (indirect crop channel)	(0.076)	(0.080)	(0.083)	(0.122)	(0.056)
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	19340	18318	21048	15198	19846
F-Stat on Direct Effect	42.4	38.3	34.1	34.1	45.0

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. All variables are in standardized units. *Male Migrants* is number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with number of baseline working age males. *Village emigration* is the leave-one-out average number of working age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants across households in the state weighted by inverse euclidean distance between actual crop portfolios between household i and i'. Columns 1 and 2 are indices for agricultural expenses and stock of machinery, respectively. Column 3 is farm size, measured as cultivated acres. Labor (column 4) is person-days of labor (household + hired workers). Farm profit (column 5) is farm income net of expenses. Standard errors clustered by district. All columns control for origin income, the uninteracted shift, drought, temperature, and rain. Standard errors clustered by district.

Table A18: Robustness of GE estimates: Nationwide Crop Markets

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants	-1.094***	-0.752***	-1.265***	-1.976***	-0.989***
(direct labour channel)	(0.240)	(0.222)	(0.266)	(0.333)	(0.200)
Village emigration	0.278***	0.193***	0.307***	0.487***	0.199***
(indirect land channel)	(0.062)	(0.056)	(0.063)	(0.086)	(0.051)
Crop region	0.639**	0.338	1.248***	0.954**	0.661***
emigration (indirect crop channel)	(0.265)	(0.241)	(0.375)	(0.483)	(0.179)
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	25924	24966	29342	20906	27458
F-Stat on Direct Effect	60.4	57.2	50.2	55.7	65.1

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. All variables are in standardized units. *Male Migrants* is number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with number of baseline working age males. *Village emigration* is the leave-one-out average number of working age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants across India weighted by crop suitability (see Section 6.1 for details). Columns 1 and 2 are indices for agricultural expenses and stock of machinery, respectively. Column 3 is farm size, measured as cultivated acres. Labor (column 4) is person-days of labor (household + hired workers). Farm profit (column 5) is farm income net of expenses. Standard errors clustered by district. All columns control for origin income, the uninteracted shift, drought, temperature, and rain. Standard errors clustered by district.

Table A19: Robustness of GE estimates: Direct and Indirect Effects (Extensive Margin)

	(1) Technology Expenses	(2) Machinery Ownership	(3) Cultivated Acres	(4) Labor Days	(5) Farm Profit
Male Migrants	-1.407***	-1.017***	-1.286***	-2.122***	-1.172***
(direct labour channel)	(0.278)	(0.251)	(0.234)	(0.314)	(0.220)
Village emigration	0.271***	0.202***	0.253***	0.383***	0.198***
(indirect land channel)	(0.054)	(0.048)	(0.043)	(0.062)	(0.042)
Crop region	0.180***	0.038	0.194***	0.297***	0.151***
emigration (indirect crop channel)	(0.049)	(0.047)	(0.040)	(0.055)	(0.035)
HH FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	69744	68686	75320	63958	72748
F-Stat on Direct Effect	71.0	69.4	65.6	64.9	74.3

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. Estimation sample includes households that owned land in both periods as well as those who did not. All variables are in standardized units. *Male Migrants* is the number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with number of baseline working age males. *Village emigration* is the leave-one-out average number of working age male migrants in the village. *Crop region emigration* is the leave-one-out number of migrants *within the state* weighted by crop suitability. Columns 1 and 2 are indices for agricultural expenses and stock of machinery, respectively. Column 3 is farm size, measured as cultivated acres. Labor (column 4) is person-days of labor (household + hired workers). Farm profit (column 5) is farm income net of expenses. Standard errors clustered by district. All columns control for origin income, the uninteracted shift, drought, temperature, and rain. Standard errors clustered by district.

Table A20: Placebo Tests of Indirect Effects

	Crop Market Placebo		Land Market Placebo
	(1) Profit	(2) Profit	(3) Farm Size
Male Migrants (direct labour channel)	-0.958*** (0.191)	-1.166*** (0.260)	-0.818*** (0.211)
Village emigration (indirect land channel)	0.164*** (0.044)	0.211*** (0.060)	0.140** (0.059)
Crop region emigration (indirect crop channel)	0.227*** (0.051)		0.226*** (0.051)
Crop region emigration (placebo crop channel)	-0.070** (0.035)		
Crop region emigration (elasticity wt.)		0.183*** (0.049)	
Crop region emigration (placebo: inv. elasticity wt.)		0.081** (0.034)	
Village emigration (placebo land channel)			0.064 (0.053)
HH FEs	✓	✓	✓
Year FEs	✓	✓	✓
Observations	27458	19846	16116
F-Stat on Direct Effect	69.8	45.2	31.5

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. The outcome is farm profits in columns 1-2 and farm size in column 3. All variables are in standardized units. *Male Migrants* is number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with number of baseline working age males. *Village emigration* is the leave-one-out average number of working age male migrants in the village. *Crop region emigration* is the weighted, leave-one-out number of migrants *within the state*. Column 1 adds a placebo where weights on *Crop region emigration* are not inverted. In column 2, weights on *Crop region emigration* are cross-price elasticities between the main crop grown by household i and i' and the placebo measure is the inverse elasticity. Column 3 adds a placebo measured as aggregate emigration among non-agricultural households in the village. All columns control for origin income, the uninteracted shift, drought, temperature, and rainfall. Standard errors clustered by district.

Table A21: IV Estimates—Direct Effects on Land Contracts

	(1) Sharecrop (=1)	(2) Rent out (=1)	(3) Both (=1)
Male Migrants (σ)	0.057**	-0.041	-0.004
	(0.028)	(0.027)	(0.008)
Wt. Income (s_{dt})	Yes	Yes	Yes
Origin Income (inc_{dt})	Yes	Yes	Yes
HH FEs	✓	✓	✓
Year FEs	✓	✓	✓
Observations	30520	30520	30520
F-Statistic	48.6	48.6	48.6

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. The outcome in column 1 is an indicator for whether the household sharecropped out land in the past year. Column 2 is an indicator for whether they rented out land. Column 3 is an indicator for whether they did both sharecropping and renting. Male Migrants is the number of working age male migrants in the household, instrumented with inverse-distance population weighted income interacted with the number of baseline working age males in the household. "Wt. Income" is inverse-distance, population-weighted income. "Origin Income" is district-level mean per-capita income. All specifications control for drought, temperature, and rainfall. Standard errors clustered by district.

Table A22: District-Level Impacts on Land Use Type

	(1) Crop	(2) Forest	(3) Grass	(4) Wetland	(5) Bare	(6) Settlement	(7) Water
Male Migrants (σ)	0.625	0.403*	-0.015	0.015	0.065*	-1.037	-0.056
	(0.712)	(0.228)	(0.137)	(0.014)	(0.037)	(0.764)	(0.146)
Origin Income (inc_{dt})	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District FEs	✓	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓	✓
Observations	718	718	718	718	718	718	718
F-Statistic	5.7	5.7	5.7	5.7	5.7	5.7	5.7

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. The outcome is percentage (0-100) of district land area under each land use type. Male migrants is district-level mean per capita number of working-age male migrants, instrumented with inverse-distance population-weighted income. "Origin income" is mean per capita income in the household's own district. Standard errors clustered by district.

Table A23: OLS Estimates—Distance to Nearest City, Agricultural Laborers, and Yields

	(1) % Δ Labor/ha	(2) % Δ Revenue/ha
Distance to Nearest City (km)	0.007	0.030*
	(0.006)	(0.016)
State FEs	✓	✓
Observations	481	483
R^2	0.290	0.309

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. This table is the empirical analog of the map in Figure 4A and 4B. Distance to the nearest city is from the centroid of each district to the nearest of the 10 high-growth cities shown in Figure 4. In column 1, the outcome is percent change in the number of agricultural laborers per ha. between 2001 and 2011. In column 2, it is percent change in crop revenue per ha, measured as the value of all crops divided by cropped area. Standard errors are heteroskedasticity robust.

B Appendix Figures

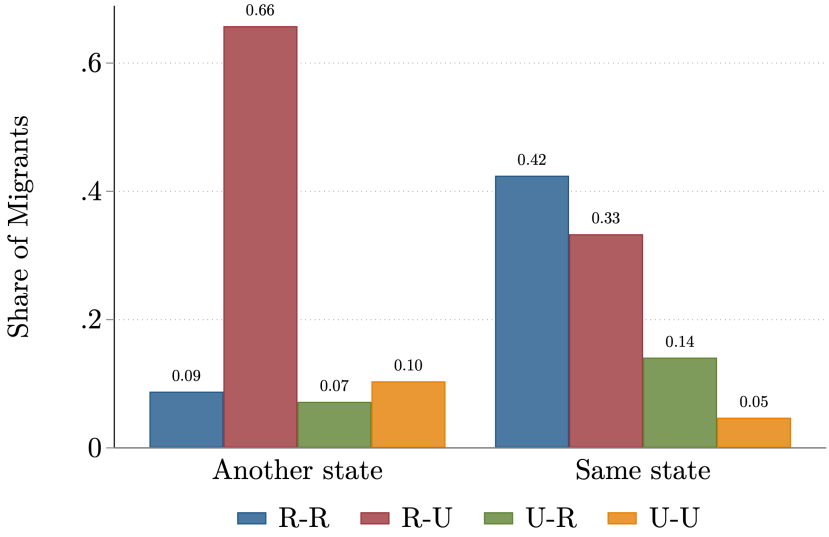


Figure B1: Migration Streams by Distance

Note: Data is at the migrant level. "R-R" denotes migrants with rural origin and rural destination, "R-U" denotes rural origin and urban destination, and so on.

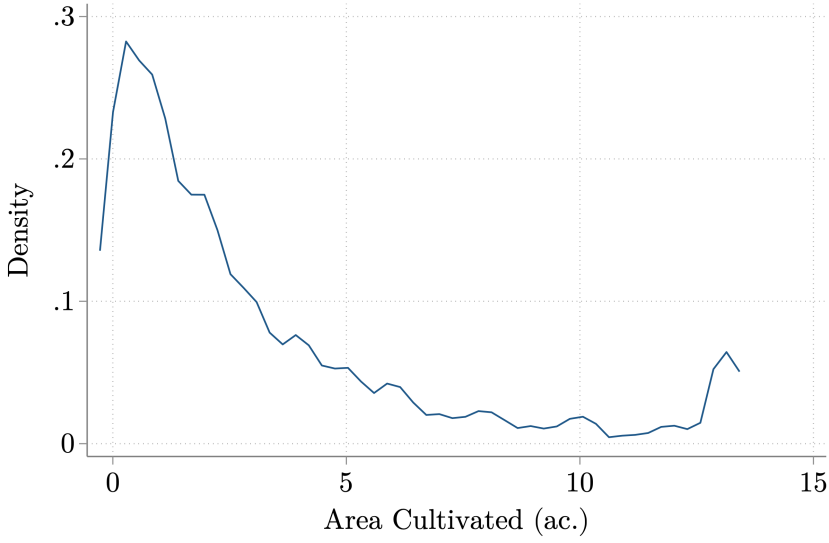


Figure B2: Distribution of Household Farm Size

Note: Kernel density plot of household farm size (cultivated area).

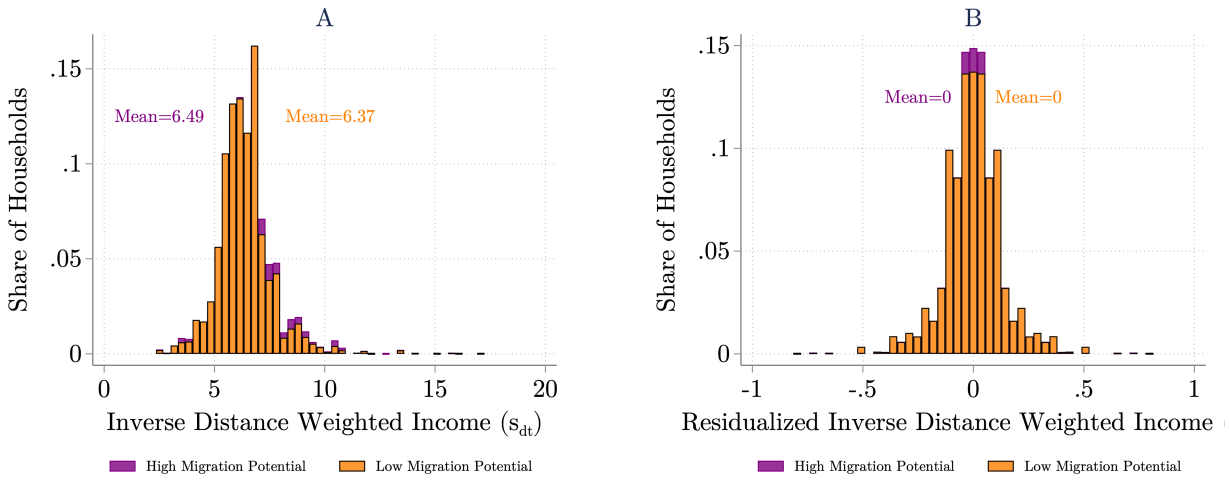


Figure B3: Distribution of Shocks

Note: Figures show the distribution of inverse-distance weighted income shocks (s_{dt}) for households with below median number males living at home during baseline (low migration potential) and above-median for the same (high migration potential). Panel A plots the raw data. Panel B plots values residualized on household and year fixed effects.

Online Theory Appendix

C Conceptual Framework

This appendix develops a model of rural production and migration with market spillovers. We use the model predictions to motivate our estimation framework, including the instrument and spillover estimation. Although the model is adapted to our context, it builds on general spatial trade models (see Allen and Arkolakis (2023) for a recent overview).

C.1 Set-up

Environment: Consider an economy that comprises many villages indexed by $j \in [1, ..., J]$ and one urban center. Villages produce crops, while urban centers produce services. Each village is endowed with fixed land A_j and is inhabited by heterogeneous households. These households draw their crop mix from a vector of crops $k \in [1, ..., K]$ based on local crop suitability, ω_{jk}. Crop suitability is homogeneous within villages and uncorrelated with distance to the urban center. While this is a simplifying assumption, none of our results is driven by it. Multiple villages can be equally suitable for the same crops. The urban center uses labor from surrounding rural areas to produce service goods. Space is characterized by distance to the urban center, which determines the opportunity cost of agricultural labor. Residents of remote villages face lower opportunity costs of agriculture than those in villages close to the urban center.

Markets: Rural labor markets are absent, while land, product, and technology markets are frictionless. We recognize that this is a strong abstraction from reality. Although none of our results hinge critically on this assumption, it allows us to make clear predictions. The labor market assumption further draws on the evidence of malfunctioning rural labor markets in India (Foster and Rosenzweig, 2022; Breza et al., 2021). The simplifying assumption of frictionless land, product, and technology markets allows us to focus on the impact of rural-to-urban migration on agriculture while abstracting from other market imperfections. Frictionless product markets equalize the crop price, p_k, across villages, similar to the assumptions in Borusyak et al. (2022b).[26] Crop prices are endogenously determined by aggregate supply within the economy.

We use the term land in a wider sense to include other immobile production factors

[26] This assumption contradicts the agricultural trade barriers across Indian states. However, we treat each state in our empirical approach as an individual economy to reconcile the theoretical assumptions with the empirical facts.

such as irrigation water or immobile capital. Land prices, ρ_j, are endogenously determined within villages and equal the marginal productivity of land. The marginal productivity of land increases with the total village agricultural labor, L_j, such that $\frac{\partial \rho_j}{\partial L_j} > 0$. Land markets equalize the marginal productivity of land across crops and households within one village.

Labor and Migration: Households supply one unit of labor inelastically, which they allocate to agriculture and the service sector. The allocation of rural labor to urban services describes rural-to-urban migration. We treat this process as a continuous rather than a binary decision in line with the large fraction of temporary migration and the migration of individual household members instead of whole households (Fact 1, Section 3.2).

Working in the service sector involves a migration cost, τ, which increases with distance to the urban center. This is a recurring cost since migrants eventually return to their origin. Similar to Morten and Oliveira (2018) and Pellegrina and Sotelo (2021), we assume iceberg migration costs (a fraction of the wage is lost) such that farmers earn net urban wage $\frac{w}{\tau}$. We assume $\tau \geq 1$, with $\tau = 1$ describing zero-cost migration since the farmer earns the full wage.

Households differ in both labor productivity and the rural–urban productivity gap. Departing from Lagakos and Waugh (2013), where heterogeneity arises in both sectors, we assume it originates solely in the urban sector. Hence, variation in the rural-urban productivity gap reflects only differences in urban productivity. This assumption yields clear predictions about heterogeneous migration responses, which underlie our empirical strategy. Migration affects agricultural productivity; however, not through differential labor productivity in agriculture, as in Lagakos and Waugh (2013), but through differential local crop suitability and spatially heterogeneous migration responses driven by differential migration costs. Concretely, agricultural labor productivity is homogeneous across households, while urban labor productivity, φ_{ij}, varies. This generates heterogeneous responses to urban productivity shocks consistent with observed migration patterns.

Demand: We do not impose a functional form on the demand system to allow non-homothetic preferences per Engel's law. We assume downward-sloping demand for crop k that weakly increases in the total income of the economy and declines in the aggregate supply of crop k: $\partial p_k / \partial Y \geq 0 \quad \text{and} \quad \partial p_k / \partial Y_k < 0$ where Y is aggregate income and Y_k is aggregate supply of crop k. Unlike agricultural goods, demand for urban goods is perfectly elastic. This reflects India's highly restricted agricultural markets alongside an export-oriented service and manufacturing sector. The urban good is the numeraire throughout the model.

Agricultural Production: Production of crop k is an increasing and concave function of labor, land, and technology defined by:

$$y_{ijk} = \omega_{jk} f(l_{ijk}, a_{ijk}, \theta_{ijk})$$

where i indexes households, j indexes villages, and k indexes crops. l_{ijk} thus describes labor of household i in village j allocated to crop k. a_{ijk} denotes land and θ_{ijk} denotes technology in the same way. ω_{jk} is village j's agricultural suitability for crop k. We further assume that land increases the marginal productivity of labor and technology, i.e.

$$\frac{\partial^2 f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk} \partial a_{ijk}} > 0 \tag{10}$$

$$\frac{\partial^2 f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial \theta_{ijk} \partial a_{ijk}} > 0 \tag{11}$$

This assumption is built into the most common production functions, including Cobb-Douglas and CES. It also conveniently reduces the number of possible cases for the results. Unlike land, technology can be labor-saving or labor-complementary. It is labor-saving if it reduces the marginal productivity of labor and vice versa for labor-complementary technologies (Acemoglu, 2010).

Definition 1. *A technology is labor-saving if*

$$\frac{\partial^2 f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk} \partial \theta_{ijk}} < 0.$$

It is labor-complementary if

$$\frac{\partial^2 f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk} \partial \theta_{ijk}} > 0$$

To simplify the analysis further, we assume that the effect of technologies on labor is homogeneous across crops, i.e., a labor-saving technology is labor-saving for all crops, and a labor-complementary technology is labor-complementary for all crops.

Urban Production: Urban production is a constant returns-to-scale function with labor as the only input. Urban output is the numeraire such that the marginal productivity of effective urban labor is given by the wage, w. Urban productivity of households, φ_{ij}, converts household labor to effective urban labor units. Although households face the same urban wage, w, opportunity costs of agriculture are heterogeneous across households because of differences in φ_{ij}.

Household Problem: Rural household i in village j maximizes profits, taking prices as given:

$$\max_{a_{ijk},\theta_{ijk},l_{ijk}} \sum_{k=1}^{K} \left[p_k \omega_{jk} f(l_{ijk}, a_{ijk}, \theta_{ijk}) - \rho_j a_{ijk} - v\theta_{ijk}\right] + \frac{w\varphi_{ij}}{\tau_j} l_{ijw} \quad \text{s.t.} \quad \sum_{k=1}^{K} l_{ijk} + l_{ijw} = 1 \tag{12}$$

where p_k is the price of crop k, ρ_j is the land price in village j, and v is the exogenously given rental rate of capital. φ_{ij} is urban labor productivity of household i from village j, w is the urban wage, $\tau_j \geq 1$ is the migration cost, which increases with distance, and l_{ijw} is the labor allocated to urban production. Optimality requires

$$\begin{aligned}
p_k \omega_{jk} \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk}} &\geq \frac{w\varphi_{ij}}{\tau_j}, \\
p_k \omega_{jk} \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial a_{ijk}} &\geq \rho_j, \\
p_k \omega_{jk} \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial \theta_{ijk}} &\geq v,
\end{aligned}$$

where the conditions hold with equality for all crops produced by the household (all k with $l_{ijk}, a_{ijk}, \theta_{ijk} > 0$) and with inequality for all crops not produced by the household (all $k' \neq k$ with $l_{ijk'}, a_{ijk'}, \theta_{ijk'} = 0$).

C.2 Theoretical Predictions

Equipped with the model primitives, we now characterize the economy's optimal labor allocation and spatial distribution of production. We then show how urbanization affects this distribution. This section mainly provides intuition for our theoretical results. The proofs are in Appendix D.

C.2.1 Spatial Organization of the Economy

The share of agriculture in village-level production increases with remoteness. The mechanism is through declining opportunity cost of agricultural labor with distance to the urban center, $\frac{w\varphi_{ij}}{\tau_j}$. For labor complementary technologies, this also implies that more technology is used with increasing distance to the urban center. A formal proof of this

result is provided in Appendix D.1.[27]

C.2.2 Urban Productivity Shocks and the Reorganization of Agriculture

We now use the model to ask how urban productivity shocks affect the spatial organization of agriculture. Throughout this study, we assume that the urban productivity shock is exogenous to the rural sector, such as the effect of India's telecom and internet rollout on productivity in the service sector. The total effect is composed of two forces: 1) a direct effect through labor reallocation and the adjustment of the other production factors, and 2) an indirect effect through aggregate emigration and the corresponding changes in crop and land prices. To characterize these effects, we first present two predictions about the effect of urban productivity shocks on agricultural labor:

Prediction 1. *The direct effect of a positive urban productivity shock on agricultural labor is negative. The magnitude of the effect declines with the distance to the urban center, and it increases with the urban productivity of the household.*

Proof. The proof is given in Appendix D.2. □

Intuitively, increasing urban productivity raises the opportunity cost of agriculture and prompts labor reallocation. Proximity to the urban center enhances this opportunity cost effect. We will use this to motivate the first stage of our estimation strategy in Section 4.2.

In the rest of the model, we define "rural-to-urban migration" as labor reallocation from agriculture to urban production in response to positive urban productivity shocks. We also call all villages in which crop k is grown the "crop-growing region" for crop k. The next prediction describes how the effect of migration differs based on the technology type:

Prediction 2. *The direct effect of rural-to-urban migration on household-level agricultural technology use is negative for labor-complementary technologies and ambiguous for labor-saving technologies. Aggregate village emigration increases household-level agricultural technology use, everything else equal. Aggregate crop-growing region emigration, which reduces aggregate production of crop k, increases household-level technology use for crop k, everything else equal.*

Proof. The proof is given in Appendix D.3. □

[27] There are special cases with labor-saving technologies where the loss of agricultural labor close to urban centers could lead to more technology adoption and possibly also to increased agricultural production if the productivity increases from technology adoption overcompensate the labor losses.

With these two predictions, we characterize the market-driven indirect effects of urban productivity shocks on production. The indirect effects of aggregate emigration on production are independent of distance to the urban center. Instead, they depend on aggregate emigration from the crop-growing region and on aggregate emigration from the same village. These are the boundaries within which crop and land prices are determined, respectively.

Urban productivity does not directly affect land and technology. It does so only through labor reallocation and corresponding changes in crop and land prices. We, therefore, describe the impact of urban productivity shocks on crop production through labor and price adjustments alone.[28]

Prediction 3. *The response of household crop production to urban income shocks can be decomposed into*

$$\frac{dy_{ijk}}{dw} = \frac{dl_{ijk}}{dw}\phi_{1ijk} + \frac{d\rho_j}{dw}\phi_{2ijk} + \frac{dp_k}{dw}\phi_{3ijk} \tag{13}$$

where ϕ_{1ijk}, ϕ_{2ijk} and ϕ_{3ijk} are composite coefficients defined in Appendix D.4.

Proof. The proof is given in Appendix D.4. □

The key insight of Prediction 3 is that urban productivity shocks trigger a direct effect (ϕ_{1ijk}) on crop production that may or may not be offset by an indirect land (ϕ_{2ijk}) or crop price effect (ϕ_{3ijk}) induced by aggregate emigration.[29] While $\phi_{2ijk} < 0$ and $\phi_{3ijk} > 0$, the sign of ϕ_{1ijk} possibly depends on whether the technology is labor-saving or labor-complementary (see Appendix D.4). ϕ_{1ijk} is positive for labor complementary technologies. Rural-to-urban migration that reduces agricultural labor and land prices and increases crop prices due to reduced crop production can, therefore, have opposing direct and indirect effects on agricultural production. The direction and magnitude of the individual effects become an empirical question.

C.3 Testing the Model's Predictions

In the main part of this paper, we empirically test these predictions, though we do not attempt a structural estimation of the model. Exploiting the panel structure of our data,

[28] Urban productivity shocks may affect agricultural production through alternative channels. For example, rural households in developing countries are often credit-constrained (Karlan et al., 2014; Fink et al., 2020). Urban income shocks could relax these constraints and provide liquidity for technology investment (Veljanoska, 2022). We refer the reader to Rozelle et al. (1999) for a simple and intuitive model of liquidity constraints and migration. Similarly, migration could also affect rural households through investment in skills and education (Dinkelman and Mariotti, 2016; Khanna et al., 2026; Dinkelman et al., 2024).

[29] In Appendix D.4, we show further that the effect of rural-to-urban migration on total household-level agricultural production is a linear combination of the crop-household-level responses.

identification relies on changes in key variables rather than their levels. For instance, the agricultural labor loss from rural-to-urban migration is proxied by $-\frac{dl_{ijk}}{dw}$. More broadly, our empirical strategy absorbs aggregate changes over time and compares outcomes under migration with a counterfactual world absent migration but otherwise subject to the same dynamics (e.g., technological progress). For example, our estimates of migration's impact on food production are silent on whether food production is generally increasing or decreasing, since time fixed effects capture common temporal trends.

Testing Prediction 1 constitutes the first stage of our analysis. The insights from this prediction guide the construction of our migration instrument. To translate the model assumptions into empirical tests, we use Euclidean distance between the household and urban centers as a proxy for iceberg migration costs and the number of working-age male household members in the baseline period as a proxy for the household's urban productivity. The latter is justified by two observations: (i) productivity differences in agriculture between men and women are small (Agarwal and Mahesh, 2023), and (ii) men are roughly four times more likely to migrate (Section 3.1.2), likely reflecting gender-specific migration barriers.

Testing Predictions 2 and 3 builds on the decomposition of direct and indirect effects underlying Prediction 3. Whether particular technologies are labor-saving or labor-complementary is often ambiguous. We therefore present results both for individual technologies and for a composite technology index. A complication in testing Prediction 3 is that the survey does not report agricultural output directly. Instead, we use agricultural profits as the dependent variable. By the Envelope Theorem, however, these estimates are theoretically equivalent. We discuss the details of this empirical strategy in Section 4.

D Proofs

D.1 Spatial Distribution of Agriculture

Here, we show the impact of distance to the urban center on crop production. Because there are many households in each location, the individual household has no market power and takes crop and land prices as given. Therefore, the first-order conditions for

crop $k \in [1, ..., K]$ are given by

$$p_k \omega_{jk} \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk}} - \frac{w\varphi_{ij}}{\tau_j} = 0, \tag{14}$$

$$p_k \omega_{jk} \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial a_{ijk}} - \rho_j = 0 \quad \text{and} \tag{15}$$

$$p_k \omega_{jk} \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial \theta_{ijk}} - v = 0. \tag{16}$$

To simplify notation, we use subscripts in the following to denote partial derivatives.[30] We also drop crop, household and village subscripts because the conditions are identical across all crops, households and villages. Next, we fully differentiate the first-order conditions with respect to the distance to the urban center, τ:

$$\begin{aligned}
p\omega(f_{ll} l_\tau + f_{la} a_\tau + f_{l\theta} \theta_\tau) + \frac{w\varphi}{\tau^2} &= 0 \\
p\omega(f_{al} l_\tau + f_{aa} a_\tau + f_{a\theta} \theta_\tau) - \rho_\tau &= 0 \\
p\omega(f_{\theta l} l_\tau + f_{\theta a} a_\tau + f_{\theta\theta} \theta_\tau) &= 0
\end{aligned}$$

Since we have assumed no trade costs for goods, crop prices are constant across space. However, the land price changes in response to the local labor allocation decisions. Solving the equations yield

$$\begin{aligned}
l_\tau &= \left[\frac{w\varphi}{\tau^2} (f_{aa} f_{\theta\theta} - f_{\theta a}^2) + \rho_\tau (f_{la} f_{\theta\theta} - f_{l\theta} f_{\theta a}) \right] D^{-1} \\
a_\tau &= \left[\frac{w\varphi}{\tau^2} (f_{l\theta} f_{\theta a} - f_{la} f_{\theta\theta}) + \rho_\tau (f_{l\theta}^2 - f_{ll} f_{\theta\theta}) \right] D^{-1} \\
\theta_\tau &= \left[\frac{w\varphi}{\tau^2} (f_{la} f_{\theta a} - f_{l\theta} f_{aa}) + \rho_\tau (f_{ll} f_{\theta a} - f_{la} f_{l\theta}) \right] D^{-1} \\
D &:= \omega p (f_{ll} f_{\theta a}^2 - f_{ll} f_{aa} f_{\theta\theta} + f_{la}^2 f_{\theta\theta} - 2 f_{la} f_{l\theta} f_{\theta a} + f_{l\theta}^2 f_{aa}) > 0
\end{aligned}$$

The denominator, D is the determinant of the Hessian matrix times $(-\omega p)$. It is positive because of the concavity assumption of the production function. The direction of the effect is, therefore, determined by the enumerator.

The total effect of distance on agricultural inputs is composed of a direct effect and an indirect effect through the response of the land price to labor reallocation to urban production. For individual farmers with low urban productivity, the indirect effect can dominate (e.g. for $\varphi = 0$). In aggregate, the indirect effect cannot dominate because it is

[30] For example $f_l := \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk}}$ and $f_{al} := \frac{\partial^2 f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk} \partial a_{ijk}}$ etc.

the response to the direct effect.

Crop-level and household-level responses: l_τ, a_τ, and θ_τ describe the crop- and household-specific responses of agricultural inputs to distance from the urban center. The household-level response is the sum of crop-level responses. Crop-specific and household-specific results are analogous because we assume homogeneous effects of technology on labor across crops.

Labor: The direct effect of distance on agricultural labor is positive because of the concavity assumption. Less labor migrates to the urban center with increasing distance to the urban center, everything else equal. The land price increases in response. The indirect effect is, therefore, primarily negative except for special cases with strong effects of labor-saving technology i.e. $f_{la}f_{\theta\theta} > f_{l\theta}f_{\theta a}$.

Land: The amount of land per household is constant if we abstract from corner solutions of labor reallocation i.e. no complete household migration. This follows from the assumption of constant land endowments per village. However, there may be land reallocation among households as households with higher urban productivity are more affected by the distance to the urban center. To see this, set $\varphi = 0$, which reduces the direct effects of labor allocation in response to the distance to the urban center to zero.

Technology: The direct effect of distance on technology is positive except for cases with labor-saving technology when it is ambiguous. The indirect effect is negative because of the concavity assumption and the Assumptions (10) and (11) ($f_{\theta a} > 0, f_{la} > 0$).

Production: The value of total production in village j is given by the sum of the values of individual production y_{ijk} i.e.

$$Y_j = \sum_i \sum_k y_{ijk}.$$

The aggregate production changes in response to distance according to

$$\begin{aligned} \frac{dY_j}{d\tau} &= \sum_i \sum_k \frac{dy_{ijk}}{d\tau} \\ &= \sum_i \sum_k p_k \omega_{jk} \left(\frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk}} \frac{\partial l_{ijk}}{\partial \tau} + \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial a_{ijk}} \frac{\partial a_{ijk}}{\partial \tau} + \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial \theta_{ijk}} \frac{\partial \theta_{ijk}}{\partial \tau} \right) \end{aligned} \tag{17}$$

The partial derivative of production with respect to labor, land, and technology is positive. The total effect is, therefore, determined by the adjustments of labor, land, and technology to distance.

Further, note that land markets equalize the marginal productivity of land across

households and crops such that $p_k \omega_{jk} \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial a_{ijk}} = \rho_j$. We can therefore write

$$\sum_i \sum_k p_k \omega_{jk} \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial a_{ijk}} \frac{\partial a_{ijk}}{\partial \tau} = \rho_j \sum_i \sum_k \frac{\partial a_{ijk}}{\partial \tau} = 0$$

since the total village land, A_j, is constant and therefore $\sum_i \sum_k \frac{\partial a_{ijk}}{\partial \tau} = 0$.

Equation (17), therefore, simplifies to

$$\frac{dY_j}{d\tau} = \sum_i \sum_k p_k \omega_{jk} \left(\frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk}} \frac{\partial l_{ijk}}{\partial \tau} + \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial \theta_{ijk}} \frac{\partial \theta_{ijk}}{\partial \tau} \right) \tag{18}$$

For labor complementary technologies, we get

$$\frac{dY_j}{d\tau} = \sum_i \sum_k p_k \omega_{jk} \left(\frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial l_{ijk}} \frac{\partial l_{ijk}}{\partial \tau} + \frac{\partial f(l_{ijk}, a_{ijk}, \theta_{ijk})}{\partial \theta_{ijk}} \frac{\partial \theta_{ijk}}{\partial \tau} \right) > 0$$

D.2 Proof of Prediction 1

Here, we show the effect of urban productivity shocks on agricultural labor. In the following, we use subscripts to denote partial derivatives and omit household, village, and crop subscripts (see Appendix D.1). First, we fully differentiate the first-order conditions with respect to the urban wage:

$$p_w \omega f_l + p\omega(f_{ll} l_w + f_{la} a_w + f_{l\theta} \theta_w) - \frac{\varphi}{\tau} = 0, \tag{19}$$

$$p_w \omega f_a + p\omega(f_{al} l_w + f_{aa} a_w + f_{a\theta} \theta_w) - \rho_w = 0, \tag{20}$$

$$p_w \omega f_\theta + p\omega(f_{\theta l} l_w + f_{\theta a} a_w + f_{\theta\theta} \theta_w) = 0. \tag{21}$$

We then solve these equations for l_w, a_w, and θ_w:

$$
\begin{aligned}
l_w =& \{\frac{\varphi}{\tau}(f_{\theta a}^2 - f_{aa}f_{\theta\theta}) \\
&+ \rho_w(f_{la}f_{\theta\theta} - f_{l\theta}f_{\theta a}) \\
&- p_w\omega[f_l(f_{\theta a}^2 - f_{aa}f_{\theta\theta}) + f_a(f_{la}f_{\theta\theta} - f_{l\theta}f_{\theta a}) + f_\theta(f_{l\theta}f_{aa} - f_{la}f_{\theta a})]\}D^{-1} \\
\theta_w =& \{\frac{\varphi}{\tau}(f_{l\theta}f_{aa} - f_{la}f_{\theta a}) \\
&+ \rho_w(f_{ll}f_{\theta a} - f_{l\theta}f_{la}) \\
&+ p_w\omega(f_l f_{la}f_{\theta a} - f_l f_{l\theta}f_{aa} + f_a f_{la}f_{l\theta} - f_{ll}f_a f_{\theta a} - f_\theta f_{la}^2 + f_{ll}f_\theta f_{aa})\}D^{-1} \\
a_w =& -\{\frac{\varphi}{\tau}(-f_{la}f_{\theta\theta} + f_{l\theta}f_{\theta a}) \\
&+ p_w\omega(f_l f_{la}f_{\theta\theta} - f_l f_{l\theta}f_{\theta a} + f_a f_{l\theta}^2 - f_{ll}f_a f_{\theta\theta} - f_\theta f_{la}f_{l\theta} + f_{ll}f_\theta f_{\theta a}) \\
&+ \rho_w(f_{ll}f_{\theta\theta} - f_{l\theta}^2)\}D^{-1} \\
D :=& \omega p(f_{ll}f_{\theta a}^2 - f_{ll}f_{aa}f_{\theta\theta} + f_{la}^2 f_{\theta\theta} - 2f_{la}f_{l\theta}f_{\theta a} + f_{l\theta}2f_{aa}) > 0
\end{aligned}
$$

The denominator, D, is the determinant of the Hessian matrix times $(-\omega p)$. This is positive because of the concavity assumption. The numerator, therefore, determines the direction of the effect. The numerator comprises a direct and indirect effect through the product and land price channels.

Here, we focus on l_w and report the effects of urban productivity on land and technology for the remaining proofs. The direct effect of urban wages on agricultural labor is given by $\frac{\varphi}{\tau}(f_{\theta a}^2 - f_{aa}f_{\theta\theta}) < 0$. It is negative because the production function is concave. Partial differentiation shows that it increases with the urban productivity of the household and declines with the distance to the urban center. The results are for crop-household-level agricultural labor. Changes in household-level agricultural labor are the sum of changes in crop-household-level agricultural labor. The results directly apply to household-level agricultural labor because the direction of crop-specific labor responses to urban productivity shocks is homogeneous across crops.

D.3 Proof of Prediction 2

Here, we show how agricultural technology use responds to labor reallocation. To do so, we express the changes in technology and land as a function of labor adjustments. We solve the first and the last of the fully differentiated first-order condition (see Appendix

D.2) for a_w and θ_w as a function of l_w :

$$\begin{aligned} a_w =& (l_w \omega p (f_{l\theta} f_{\theta a} - f_{la} f_{\theta\theta}) + p_w \omega (f_\theta f_{\theta a} - f_a f_{\theta\theta}) + \rho_w f_{\theta\theta}) D_2^{-1} \\ \theta_w =& (l_w \omega p (f_{la} f_{\theta a} - f_{l\theta} f_{aa}) + p_w \omega (f_a f_{\theta a} - f_\theta f_{aa}) - \rho_w f_{\theta a}) D_2^{-1} \\ D_2 :=& (f_{aa} f_{\theta\theta} - f_{\theta a}^2) \omega p > 0 \end{aligned}$$

D_2 is the determinant of the second-order minor. Here, we focus on θ_w and report the result for a_w only to support the proof in Appendix D.4. The direct effect is given by $l_w \omega p (f_{l\theta} f_{\theta a} - f_{la} f_{\theta\theta})$. The first part, $l_w \omega p$, is negative (see Prediction 1). The second part is ambiguous. To see this, note that $f_{la} f_{\theta\theta} < 0$ because of the concavity assumption and assumption (10). Further, note that $f_{l\theta} f_{\theta a} > 0$ for $f_{l\theta} > 0$ and $f_{l\theta} f_{\theta a} < 0$ for $f_{l\theta} < 0$ because of assumption (11).

Define household-level technology use as $\bar{\theta}_{ij} := \sum_{k=1}^{K} \theta_{ijk}$ and therefore $\frac{d\bar{\theta}_{ij}}{dw} = \sum_{k=1}^{K} \frac{d\theta_{ijk}}{d_w}$. If $\frac{d\theta_{ijk}}{d_w} > (<) 0$, then also $\frac{d\bar{\theta}_{ij}}{dw} > (<) 0$ because of the assumption that the effect of technology on labor is homogeneous across crops i.e., the direction of the technology response to urban productivity shocks is the same for all crops. The same argument applies to the remainder of the proof.

The effect of village emigration on technology use is determined by $-\rho_w f_{\theta a}$. Aggregate village emigration reduces land prices by assumption (see Section C) such that $\rho_w < 0$. Assumption (11) completes the proof for the second statement of Prediction 2.

Aggregate crop-growing region emigration, which reduces aggregate production of crop k, increases the crop price by assumption (see Section C) such that $p_w > 0$. The direction of this indirect effect is therefore determined by $\omega(f_a f_{\theta a} - f_\theta f_{aa})$. $f_a f_{\theta a} > 0$ by assumption (positive marginal productivity and assumption (11)) while $f_\theta f_{aa} < 0$ by assumption (positive marginal productivity and concavity) which completes the proof for the last statement of Prediction 2.

D.4 Proof of Prediction 3

Here, we decompose the impact of urban productivity shocks on household crop production. We use subscripts to denote partial derivatives instead of crop, household, and village indices. Using the results and definitions from Appendix D.3 we can express the

response of household crop production to urban productivity shocks as

$$\begin{aligned}
y_w =& f_l l_w + f_a a_w + f_\theta \theta_w \\
=& f_l l_w \\
&+ f_a(l_w \omega p(f_{l\theta} f_{\theta a} - f_{la} f_{\theta\theta}) + p_w \omega(f_\theta f_{\theta a} - f_a f_{\theta\theta}) + \rho_w f_{\theta\theta}) D_2^{-1} \\
&+ f_\theta(l_w \omega p(f_{la} f_{\theta a} - f_{l\theta} f_{aa}) + p_w \omega(f_a f_{\theta a} - f_\theta f_{aa}) - \rho_w f_{\theta a}) D_2^{-1} \\
=& l_w \left[f_l + \frac{\omega p(f_{l\theta} f_{\theta a} f_a - f_{la} f_{\theta\theta} f_a + f_{la} f_{\theta a} f_\theta - f_{l\theta} f_{aa} f_\theta)}{D_2} \right] \\
&+ \rho_w \frac{f_{\theta\theta} - f_{\theta a}}{D_2} \\
&+ p_w \frac{\omega(f_\theta f_{\theta a} - f_a f_{\theta\theta} + f_a f_{\theta a} - f_\theta f_{aa})}{D_2}
\end{aligned}$$

Define

$$\begin{aligned}
\phi_{1ijk} &:= \left[f_l + \frac{\omega p(f_{l\theta} f_{\theta a} f_a - f_{la} f_{\theta\theta} f_a + f_{la} f_{\theta a} f_\theta - f_{l\theta} f_{aa} f_\theta)}{D_2} \right] \\
\phi_{2ijk} &:= \frac{f_{\theta\theta} - f_{\theta a}}{D_2} < 0 \\
\phi_{3ijk} &:= \frac{\omega(f_\theta f_{\theta a} - f_a f_{\theta\theta} + f_a f_{\theta a} - f_\theta f_{aa})}{D_2} > 0.
\end{aligned}$$

These composite parameters are unaffected by marginal changes in urban productivity. We added subscripts to underline that they are household-crop-specific. For labor complementary technologies, ϕ_{1ijk} is positive, while it is ambiguous for labor-saving technologies.

The results are for crop-household combinations. To derive household-level results, we reintroduce household, village, and crop subscripts. Total household agricultural production is given by

$$\bar{y}_{ij} = \sum_{k=1}^{K} y_{ijk}$$

and the response of household production to urban productivity shocks by

$$\frac{d\bar{y}_{ij}}{dw} = \sum_{k=1}^{K} \frac{dy_{ijk}}{dw} = \sum_{k=1}^{K} \frac{dl_{ijk}}{dw} \phi_{1ijk} + \frac{d\rho_j}{dw} \sum_{k=1}^{K} \phi_{2ijk} + \sum_{k=1}^{K} \frac{dp_k}{dw} \phi_{3ijk}.$$